

# Model Analysis

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# 1 Data analysis /Dataframe import and filters

## 1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                             feats_to_balance + feats_grouping +
                             feats_reference+ feats_postproc))

    df = df[cols]
```

```
#df['campaign'] = "ALL"
```

```
return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

## 1.2 Metadata

|                             |      |
|-----------------------------|------|
| Nombre de lignes df         | 8168 |
| Nombre de lignes df clean   | 8168 |
| Nombre de features          | 5    |
| % de NaN                    | 0.0  |
| Taille dataset entraînement | 5717 |
| Taille dataset test         | 2450 |
| Métrique utilisée           | r2   |

**Column target :** i\_NH3

**Features\_X :** r\_NH3, r\_temperature, r\_umidity, r\_CO2, r\_THI

**feats\_to\_balance :**

**feats\_grouping :**

**feats\_postproc :** datetime, id\_channel, id\_rabbit, campaign

**feats\_ref :**

**id\_cols :** campaign

## 2 Model(s) analysis

### 2.1 Explanation legend tables

#### Columns

- rank: ranking regarding cross validation test
- train\_cv: mean score train of the CV
- test\_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test\_cv
- r\_test: test/test\_cv
- r\_train: train\_cv/test\_cv

#### Colors:

[Orange]  $r\_test < 0.95$  (under performance of the test)

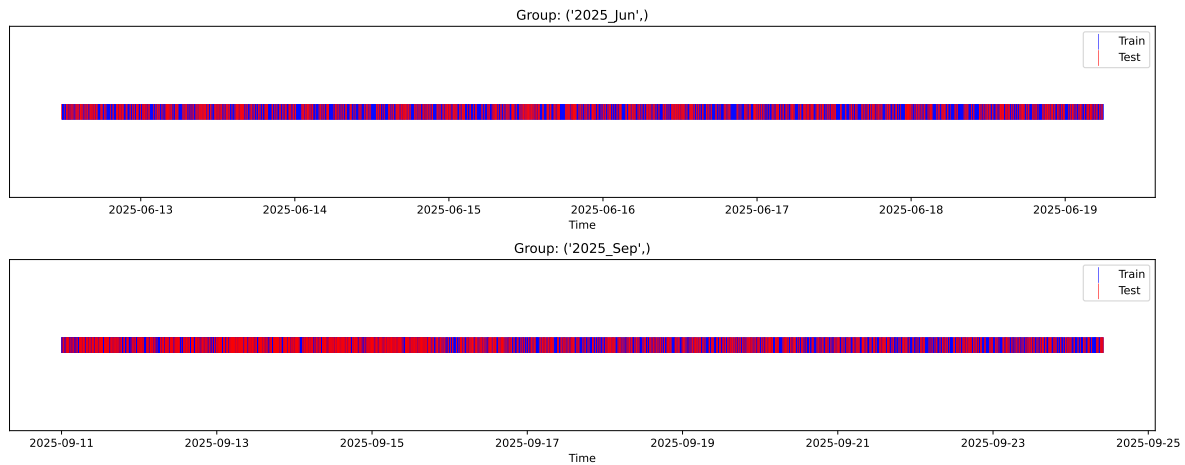
[Green]  $r\_test > 1.05$  (over performance of the test)

[Red]  $r\_train$  not in  $[0.95, 1.05]$  (gap between train/test)

## 2.2 Repartition of train and test

### 2.2.1 External Split: $X_{\text{train}}$ vs $X_{\text{test}}$

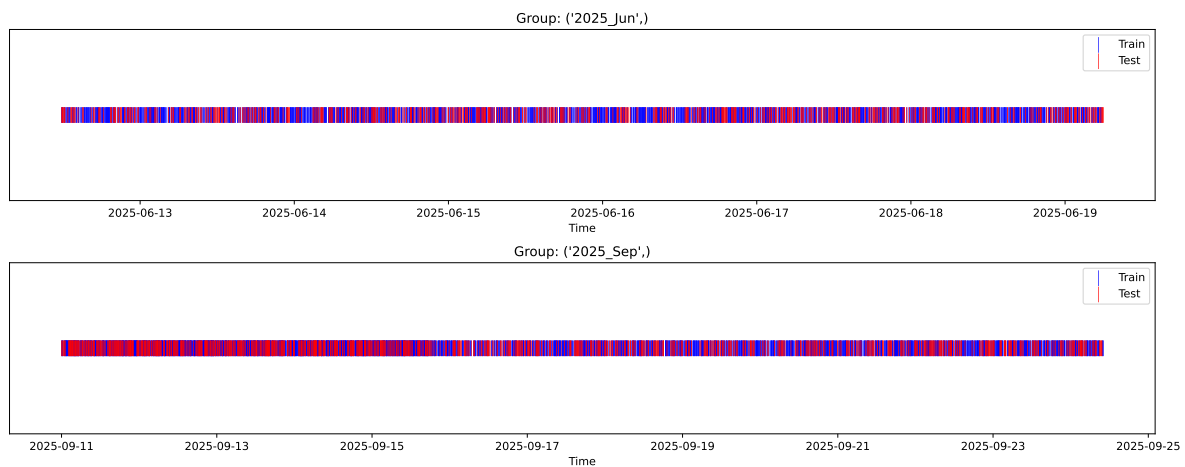
External Split: Train: 5717 samples Test: 2451 samples



### 2.2.2 Internal CV Folds

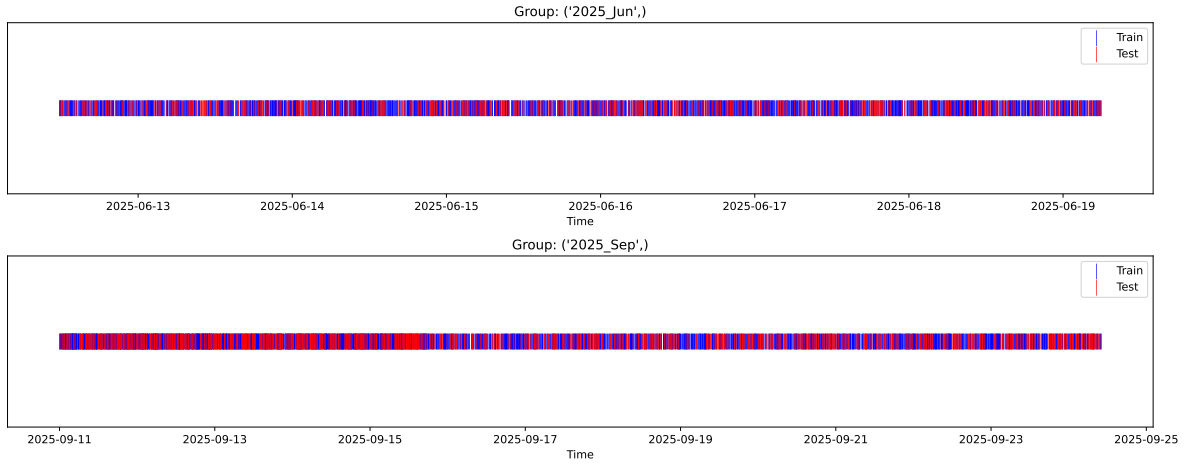
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



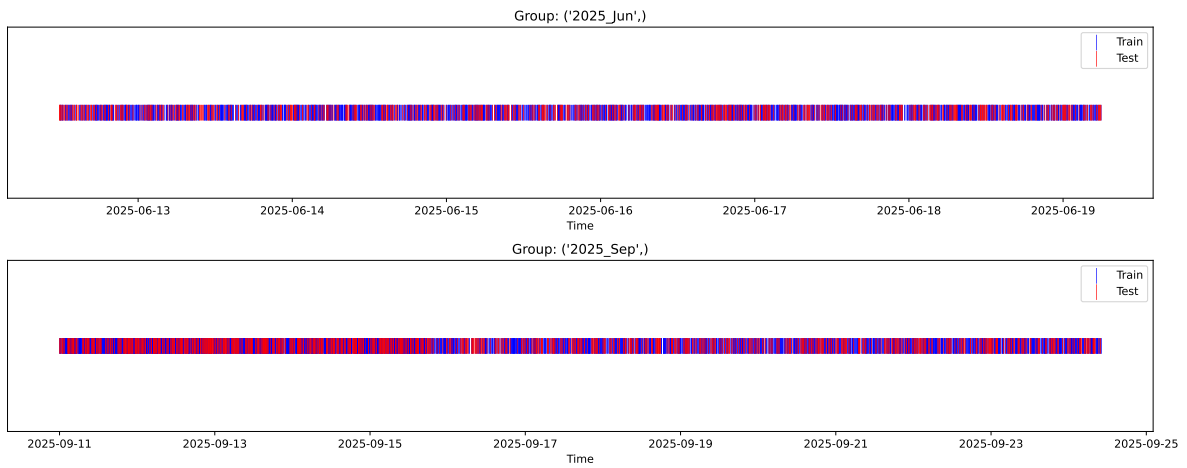
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



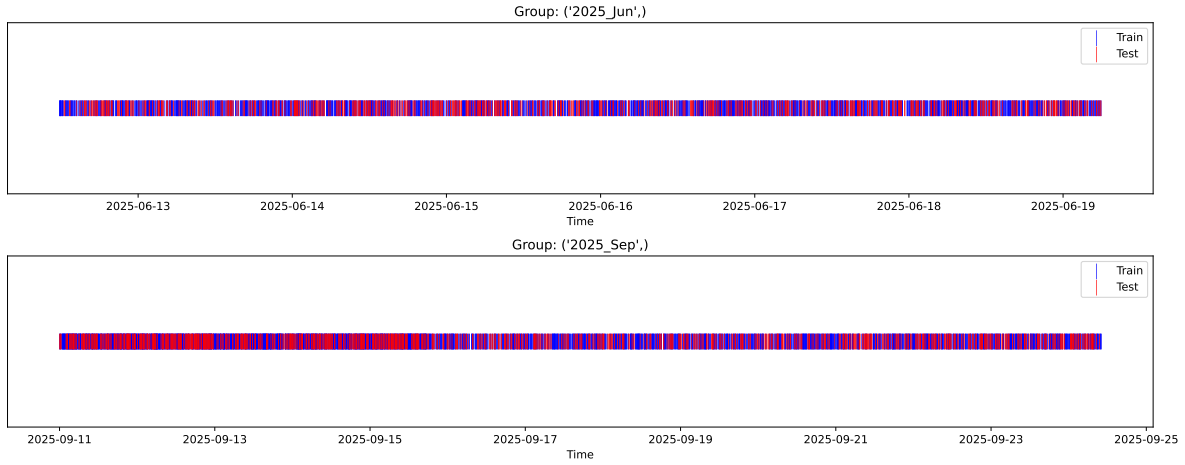
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



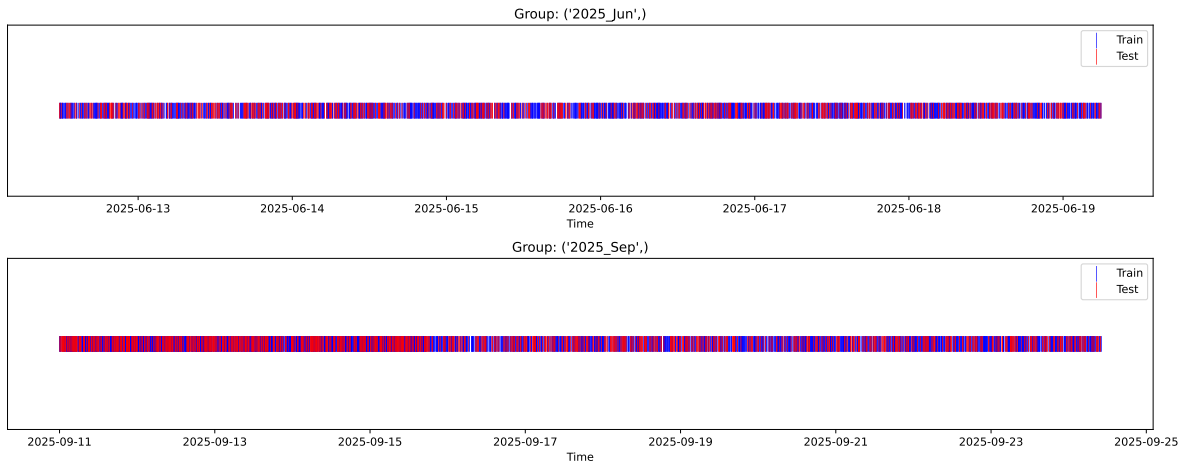
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



## 2.3 Model : Linear

### 2.3.1 Gridsearch

Distribution  $y_{\text{train}}$  vs  $y_{\text{test}}$ :  $y_{\text{train}}$ : mean=2.339, std=0.798, min=0.959, max=13.845  
 $y_{\text{test}}$ : mean=2.374, std=0.786, min=1.071, max=7.277

Fold 0 —  $y_{\text{test}}$ : mean=2.34, std=0.82,  $y_{\text{train}}$ : mean=2.34, std=0.79

Fold 1 —  $y_{\text{test}}$ : mean=2.34, std=0.77,  $y_{\text{train}}$ : mean=2.34, std=0.81

Fold 2 —  $y_{\text{test}}$ : mean=2.36, std=0.85,  $y_{\text{train}}$ : mean=2.33, std=0.78



Fold 3 — y\_test: mean=2.36, std=0.84, y\_train: mean=2.33, std=0.78

Fold 4 — y\_test: mean=2.35, std=0.76, y\_train: mean=2.34, std=0.81

train scores CV: [0.35126041 0.34129782 0.38941971 0.36572649 0.35147845]

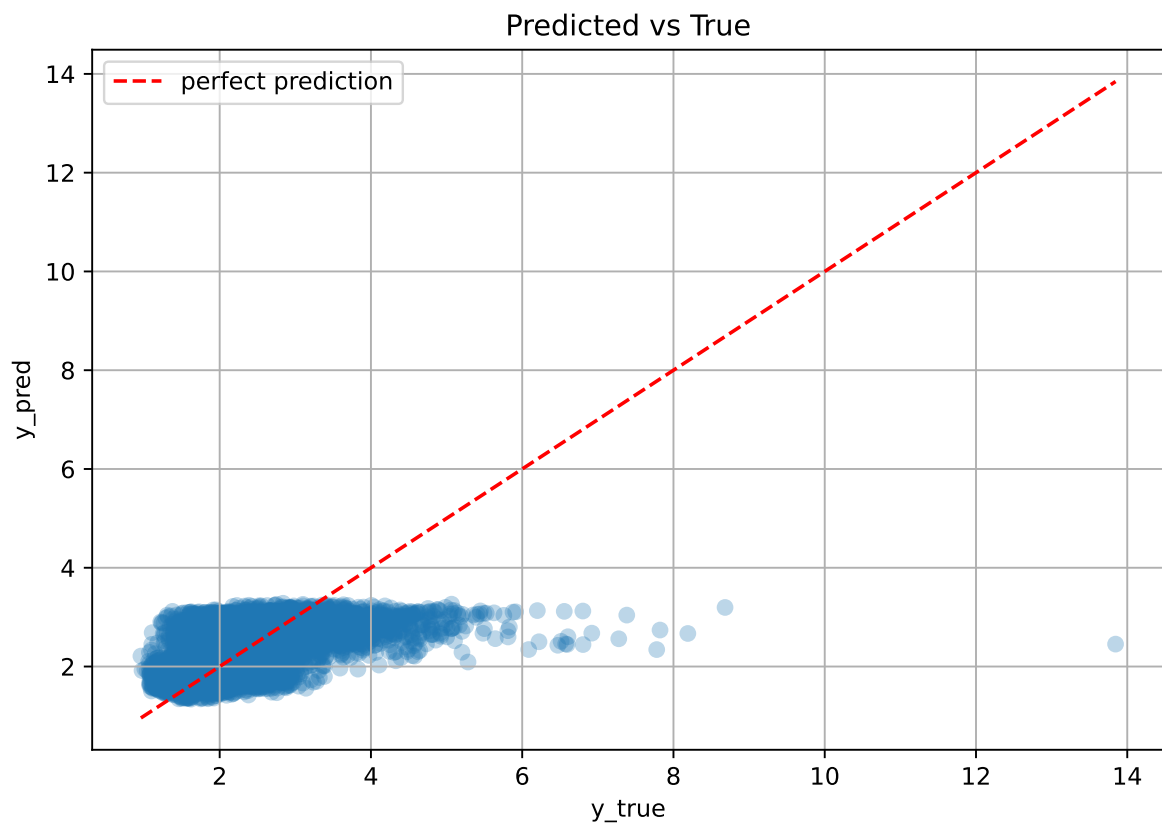
test scores CV: [0.37881413 0.40998904 0.30281638 0.34887505 0.38406497]

| rank | mean_train_cv | mean_test_cv | std_test_cv | test_score_cv | test   | diff    | r_test | r_train |
|------|---------------|--------------|-------------|---------------|--------|---------|--------|---------|
| 1.0  | 0.3598        | 0.3649       | 0.0366      | 0.3451        | 0.3453 | -0.0197 | 0.9461 | 0.9861  |

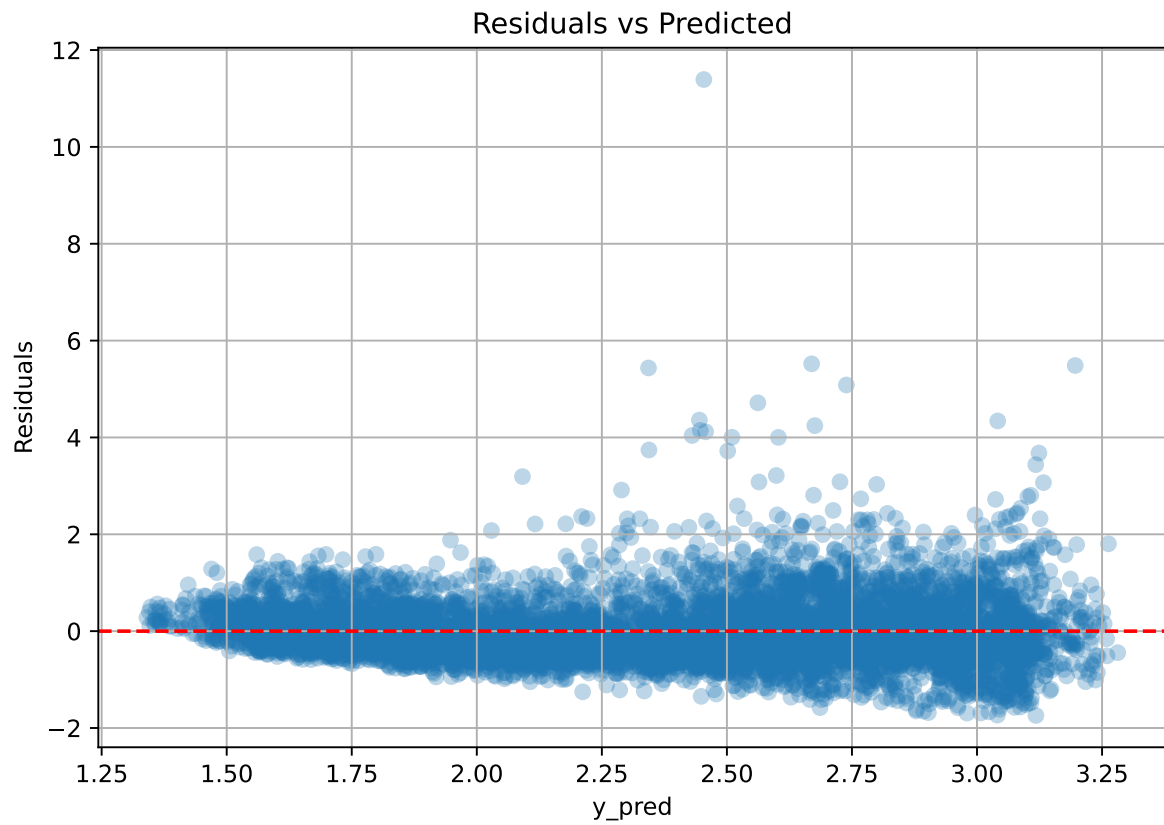
**Coefficients:**

coef\_r\_CO2 = -0.0491, coef\_r\_NH3 = -0.0447, coef\_r\_THI = 0.5970, coef\_r\_temperature = -0.7372, coef\_r\_umidity = -0.6671, intercept = 2.3392,

### 2.3.2 Scatter



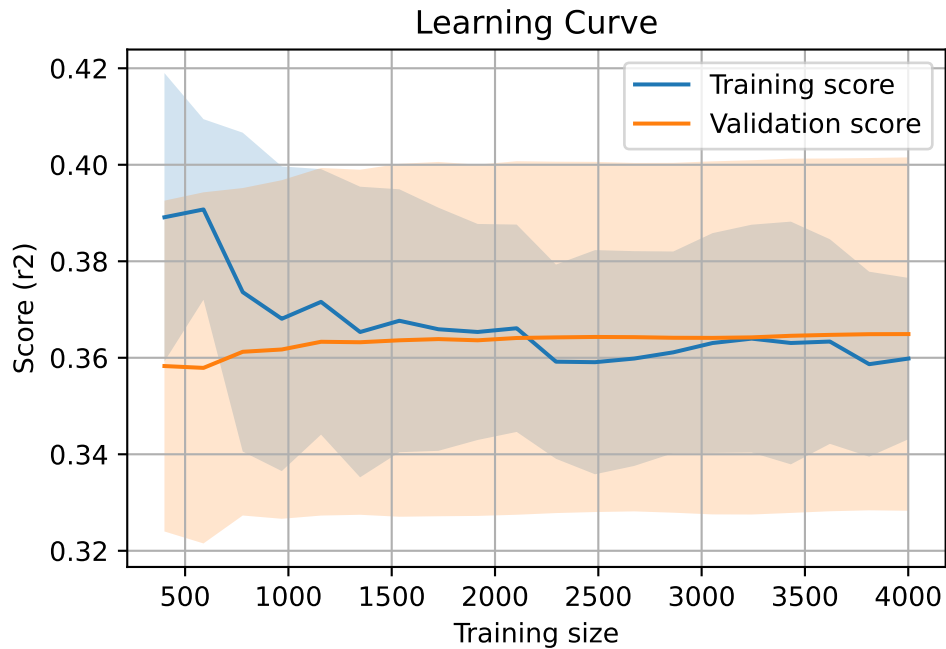
### 2.3.3 Scatter residuals



### 2.3.4 Learning curve

**Len(group)** : 8168, **Len(training)** : 5718,

**Len(training\_real)** : 4003, **Len(test\_of\_training)** : 1715, **Len(test)** : 2450

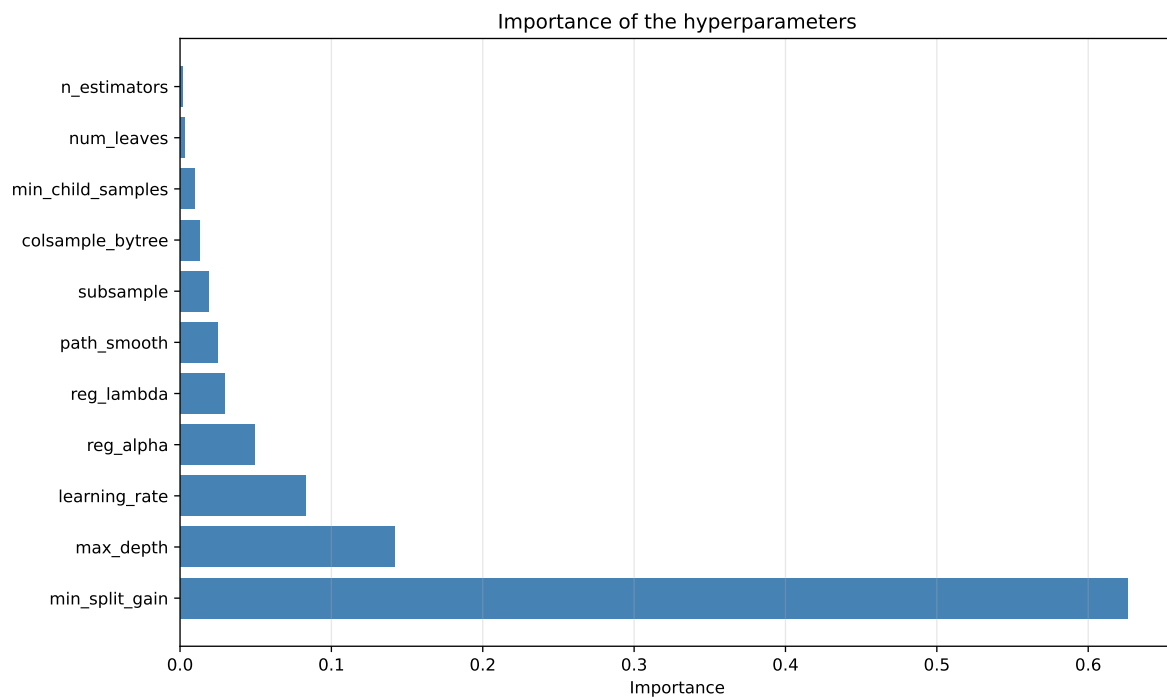
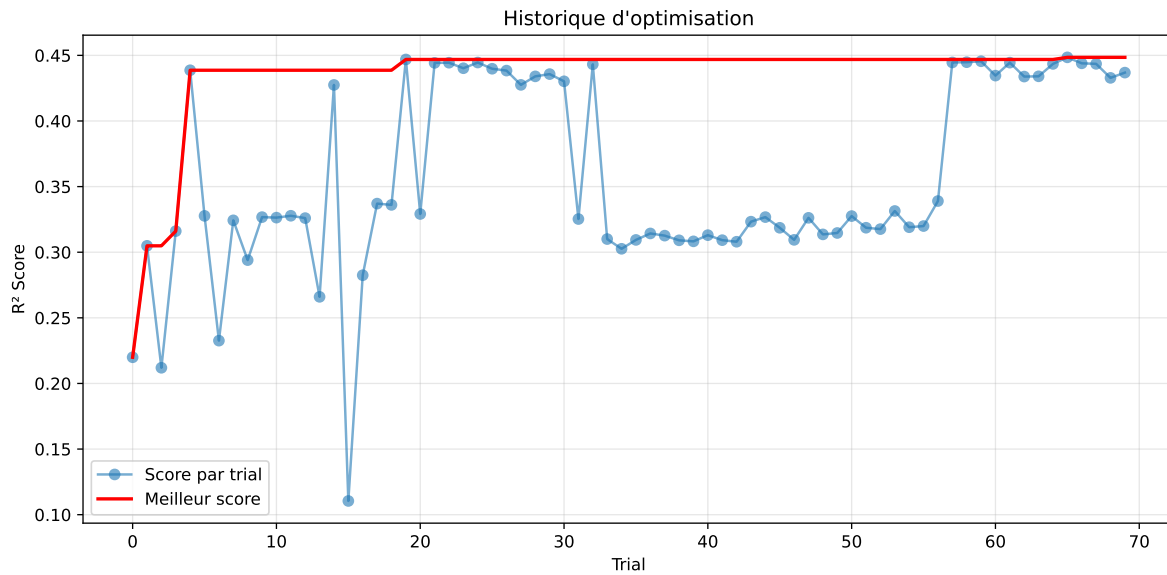


## 2.4 Model : LGBM

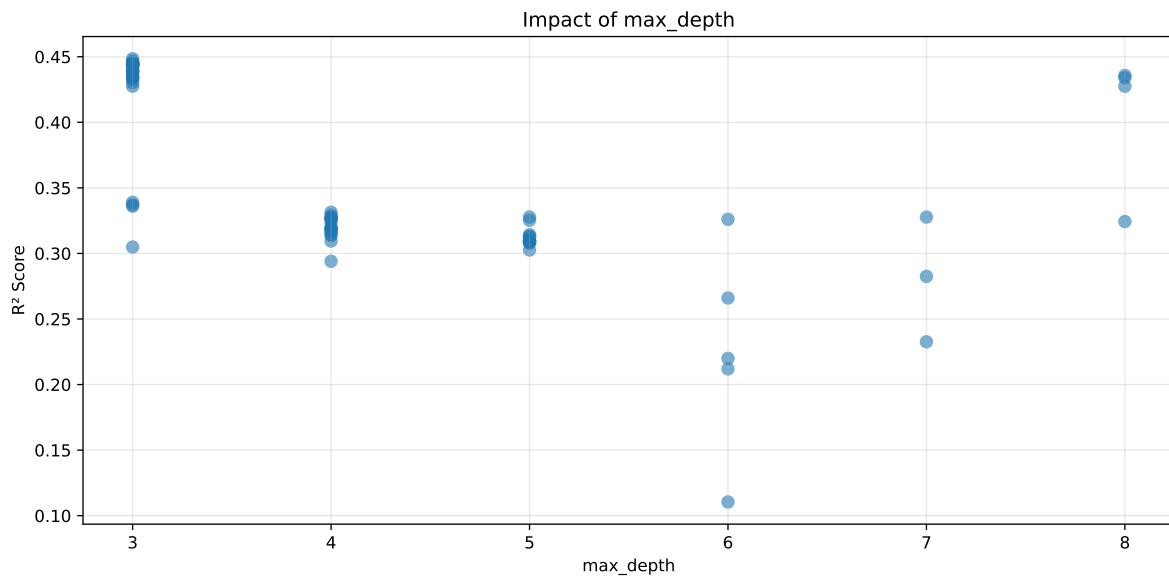
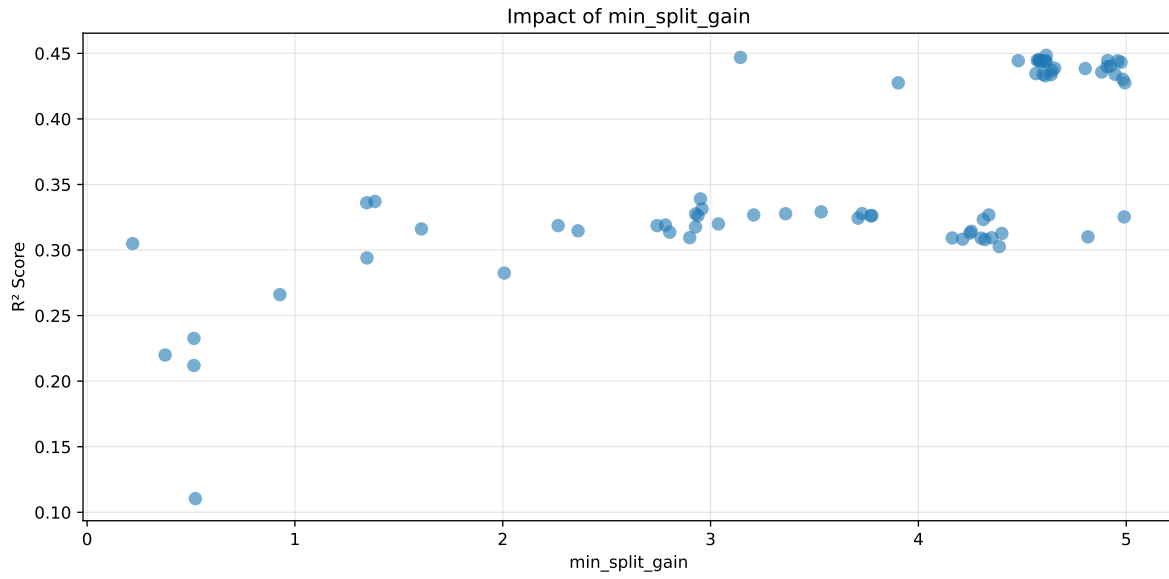
### 2.4.1 Gridsearch

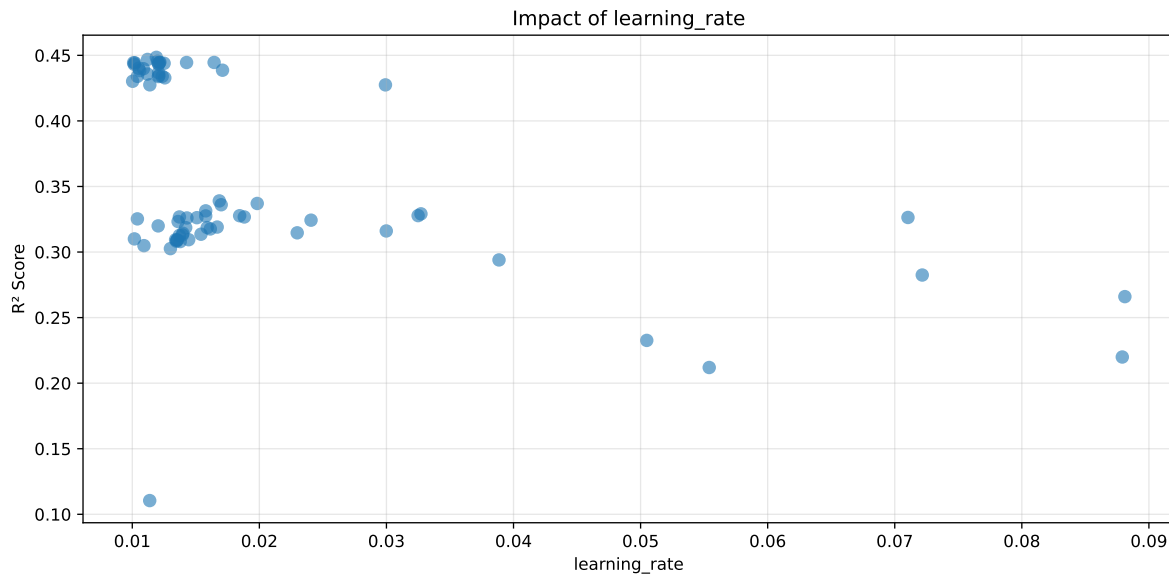
Distribution y\_train vs y\_test: y\_train: mean=2.339, std=0.798, min=0.959, max=13.845  
y\_test: mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====  
feature importance r\_umidity 491 r\_CO2 428 r\_NH3 295 r\_THI 251 r\_temperature 204



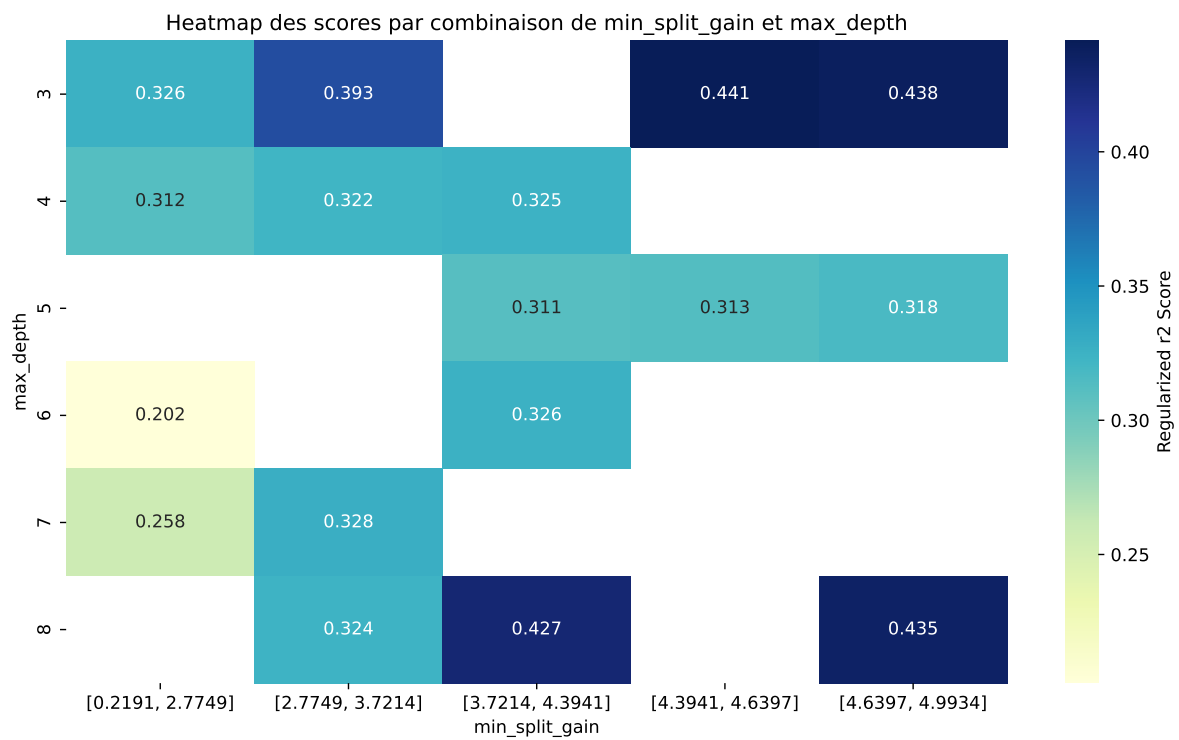
Paramètres avec >5% d'importance : ['min\_split\_gain', 'max\_depth', 'learning\_rate']





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The default value of observed=False is deprecated and will change to observed=True in a future



| rank | mean_train_cv | mean_test_cv | test_score_cv | std_test_cv | penalized_score | test   | diff   | r_test | r_train |
|------|---------------|--------------|---------------|-------------|-----------------|--------|--------|--------|---------|
| 1    | 0.4694        | 0.4485       | 0.4597        | 0.0368      | 0.4485          | 0.4752 | 0.0267 | 1.0596 | 1.0466  |
| 2    | 0.4688        | 0.4469       | 0.4584        | 0.0363      | 0.4469          | 0.4826 | 0.0357 | 1.0798 | 1.0491  |
| 3    | 0.4659        | 0.4455       | 0.4572        | 0.0374      | 0.4455          | 0.4775 | 0.032  | 1.0718 | 1.0458  |
| 4    | 0.4648        | 0.4448       | 0.4557        | 0.0375      | 0.4448          | 0.4783 | 0.0335 | 1.0754 | 1.045   |
| 5    | 0.4637        | 0.4446       | 0.455         | 0.0372      | 0.4446          | 0.4772 | 0.0326 | 1.0733 | 1.043   |
| 6    | 0.4642        | 0.4445       | 0.455         | 0.0371      | 0.4445          | 0.4741 | 0.0295 | 1.0665 | 1.0442  |
| 7    | 0.4647        | 0.4445       | 0.4556        | 0.0377      | 0.4445          | 0.4778 | 0.0333 | 1.075  | 1.0456  |
| 8    | 0.4649        | 0.4444       | 0.456         | 0.0369      | 0.4444          | 0.476  | 0.0316 | 1.071  | 1.0462  |
| 9    | 0.4639        | 0.4443       | 0.4559        | 0.0372      | 0.4443          | 0.4759 | 0.0316 | 1.0712 | 1.0441  |
| 10   | 0.4637        | 0.4439       | 0.4544        | 0.0377      | 0.4439          | 0.4745 | 0.0305 | 1.0688 | 1.0445  |
| ---  | ---           | ---          | ---           | ---         | ---             | ---    | ---    | ---    | ---     |
| 70   | 0.6095        | 0.5263       | 0.5609        | 0.0388      | 0.1105          | 0.5689 | 0.0426 | 1.081  | 1.158   |

### Hyperparameters:

Rank 1: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.011895151742815505, 'reg\_\_num\_leaves': 57, 'reg\_\_subsample': 0.7831901157782531, 'reg\_\_colsample\_bytree': 0.6381147045388039, 'reg\_\_min\_child\_samples': 13, 'reg\_\_reg\_alpha': 0.5303744228765144, 'reg\_\_reg\_lambda': 11.047712011519971, 'reg\_\_min\_split\_gain': 4.614969173309836, 'reg\_\_path\_smooth': 1.573083668133399}

Rank 2: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.01120104748788332, 'reg\_\_num\_leaves': 62, 'reg\_\_subsample': 0.8297482113290949, 'reg\_\_colsample\_bytree': 0.5779310130768743, 'reg\_\_min\_child\_samples': 10, 'reg\_\_reg\_alpha': 7.701900933392681, 'reg\_\_reg\_lambda': 4.463562603102228, 'reg\_\_min\_split\_gain': 3.143885768806488, 'reg\_\_path\_smooth': 3.324568689998213}

Rank 3: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.011986440223513103, 'reg\_\_num\_leaves': 59, 'reg\_\_subsample': 0.7678963094837117, 'reg\_\_colsample\_bytree': 0.6370202496049616, 'reg\_\_min\_child\_samples': 10, 'reg\_\_reg\_alpha': 0.6253217778778126, 'reg\_\_reg\_lambda': 18.53619906844688, 'reg\_\_min\_split\_gain': 4.583040850127482, 'reg\_\_path\_smooth': 1.5158770263093384}

Rank 4: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.01216942872275836, 'reg\_\_num\_leaves': 59, 'reg\_\_subsample': 0.7722327262312197, 'reg\_\_colsample\_bytree': 0.6494328001059522, 'reg\_\_min\_child\_samples': 10, 'reg\_\_reg\_alpha': 0.6117587516857659, 'reg\_\_reg\_lambda': 18.552759267869632, 'reg\_\_min\_split\_gain': 4.57175013206067, 'reg\_\_path\_smooth': 2.2581022946497304}

Rank 5: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.016439506415143114, 'reg\_\_num\_leaves': 59, 'reg\_\_subsample': 0.7412662087586444,



'reg\_\_colsample\_bytree': 0.6111681625590829, 'reg\_\_min\_child\_samples': 26, 'reg\_\_reg\_alpha': 0.5894169097332145, 'reg\_\_reg\_lambda': 18.901542600569154, 'reg\_\_min\_split\_gain': 4.610618241343473, 'reg\_\_path\_smooth': 0.7459515037844078}

Rank 6: {'reg\_\_n\_estimators': 600, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.014272703971301252, 'reg\_\_num\_leaves': 28, 'reg\_\_subsample': 0.5030507991702722, 'reg\_\_colsample\_bytree': 0.5449716641728058, 'reg\_\_min\_child\_samples': 18, 'reg\_\_reg\_alpha': 0.5746702162144475, 'reg\_\_reg\_lambda': 13.257679684480998, 'reg\_\_min\_split\_gain': 4.911100494356985, 'reg\_\_path\_smooth': 1.5534305682684204}

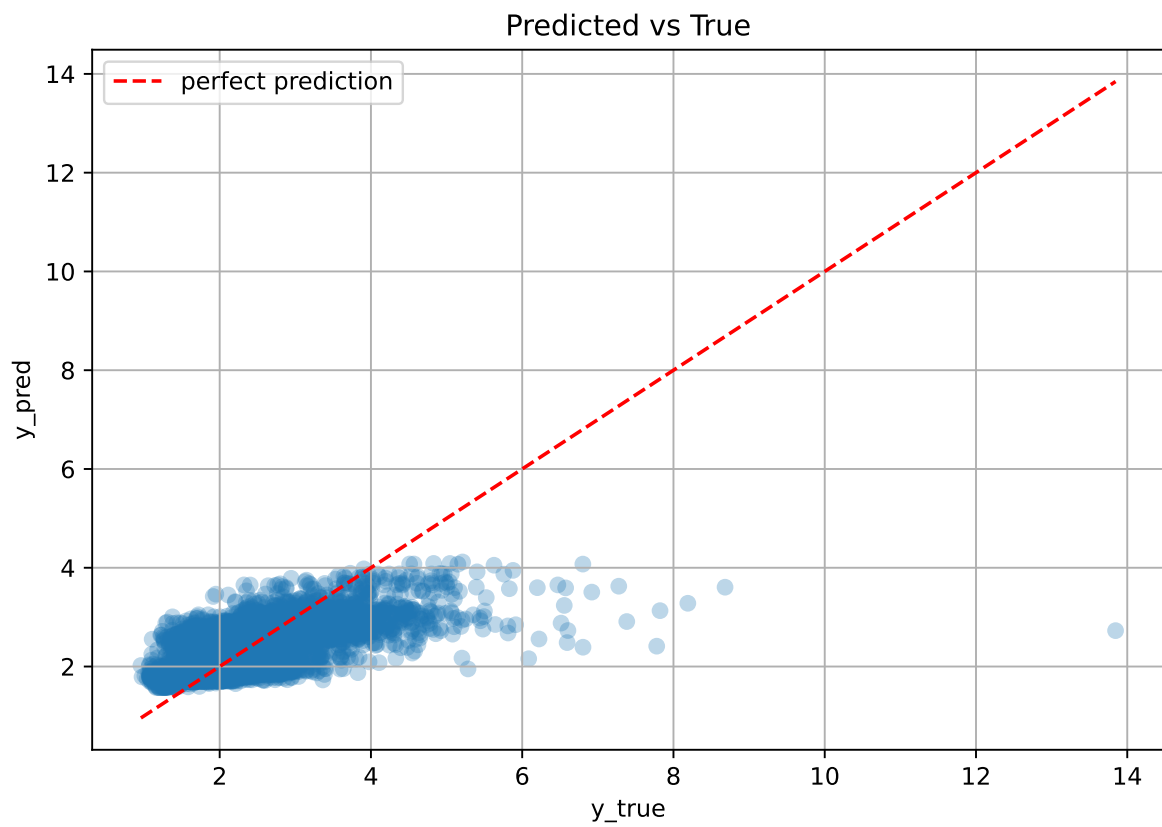
Rank 7: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.012130746539292648, 'reg\_\_num\_leaves': 59, 'reg\_\_subsample': 0.6780166486616442, 'reg\_\_colsample\_bytree': 0.6460445938266267, 'reg\_\_min\_child\_samples': 10, 'reg\_\_reg\_alpha': 0.653691773818064, 'reg\_\_reg\_lambda': 18.869642814628804, 'reg\_\_min\_split\_gain': 4.578786297429202, 'reg\_\_path\_smooth': 0.8753554214744187}

Rank 8: {'reg\_\_n\_estimators': 600, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.01010583224714535, 'reg\_\_num\_leaves': 60, 'reg\_\_subsample': 0.7439459076842614, 'reg\_\_colsample\_bytree': 0.563712332181225, 'reg\_\_min\_child\_samples': 10, 'reg\_\_reg\_alpha': 2.035866975065292, 'reg\_\_reg\_lambda': 15.012891218003624, 'reg\_\_min\_split\_gain': 4.480471283256241, 'reg\_\_path\_smooth': 1.906005382627093}

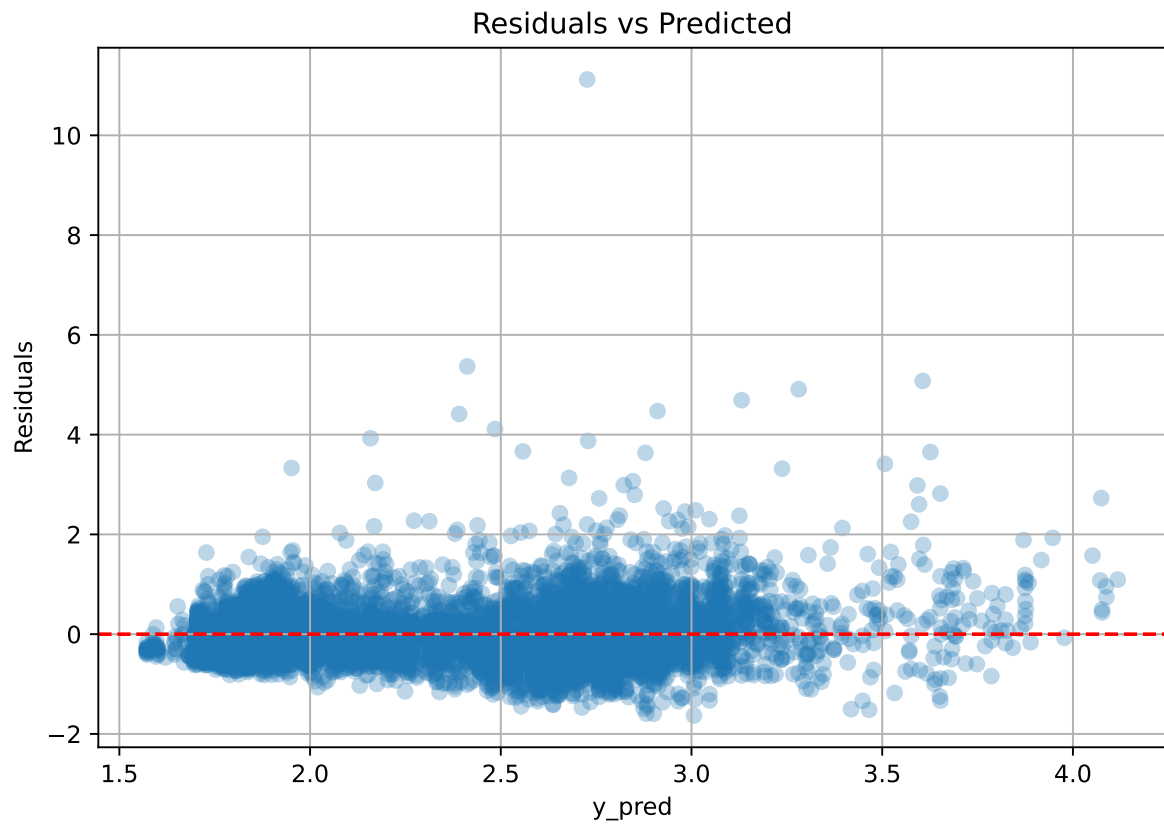
Rank 9: {'reg\_\_n\_estimators': 700, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.010205410330446562, 'reg\_\_num\_leaves': 63, 'reg\_\_subsample': 0.7496962556093668, 'reg\_\_colsample\_bytree': 0.5366940916600693, 'reg\_\_min\_child\_samples': 22, 'reg\_\_reg\_alpha': 0.20421696523131327, 'reg\_\_reg\_lambda': 13.448999736218864, 'reg\_\_min\_split\_gain': 4.959379295734436, 'reg\_\_path\_smooth': 1.564816808972107}

Rank 10: {'reg\_\_n\_estimators': 800, 'reg\_\_max\_depth': 3, 'reg\_\_learning\_rate': 0.012505262256374343, 'reg\_\_num\_leaves': 20, 'reg\_\_subsample': 0.7776649851970807, 'reg\_\_colsample\_bytree': 0.641470913547315, 'reg\_\_min\_child\_samples': 12, 'reg\_\_reg\_alpha': 2.2143262394273635, 'reg\_\_reg\_lambda': 10.444468537614654, 'reg\_\_min\_split\_gain': 4.601764798418578, 'reg\_\_path\_smooth': 2.222876145118425}

### 2.4.2 Scatter



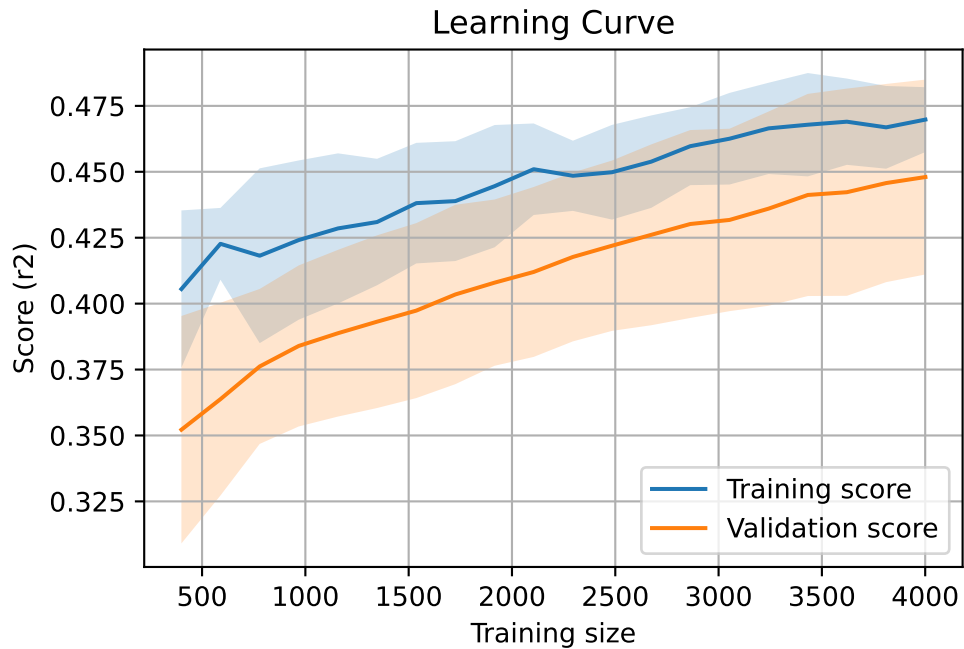
### 2.4.3 Scatter residuals



### 2.4.4 Learning curve

`Len(group)` : 8168, `Len(training)` : 5718,

`Len(training_real)` : 4003, `Len(test_of_training)` : 1715, `Len(test)` : 2450

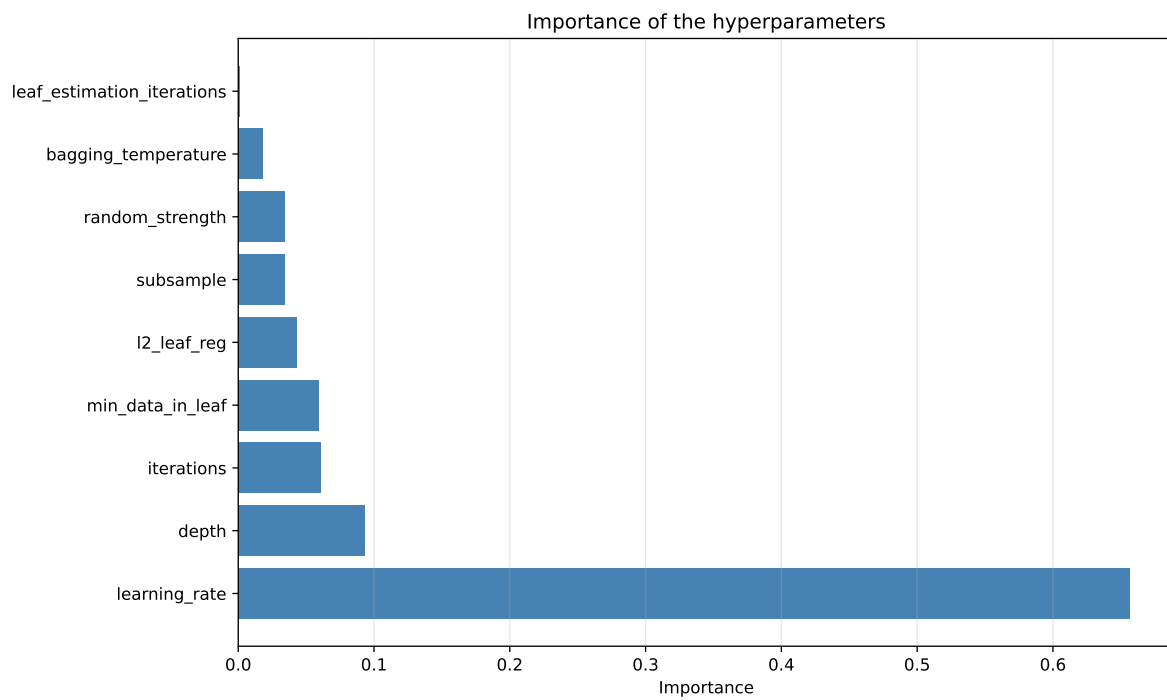
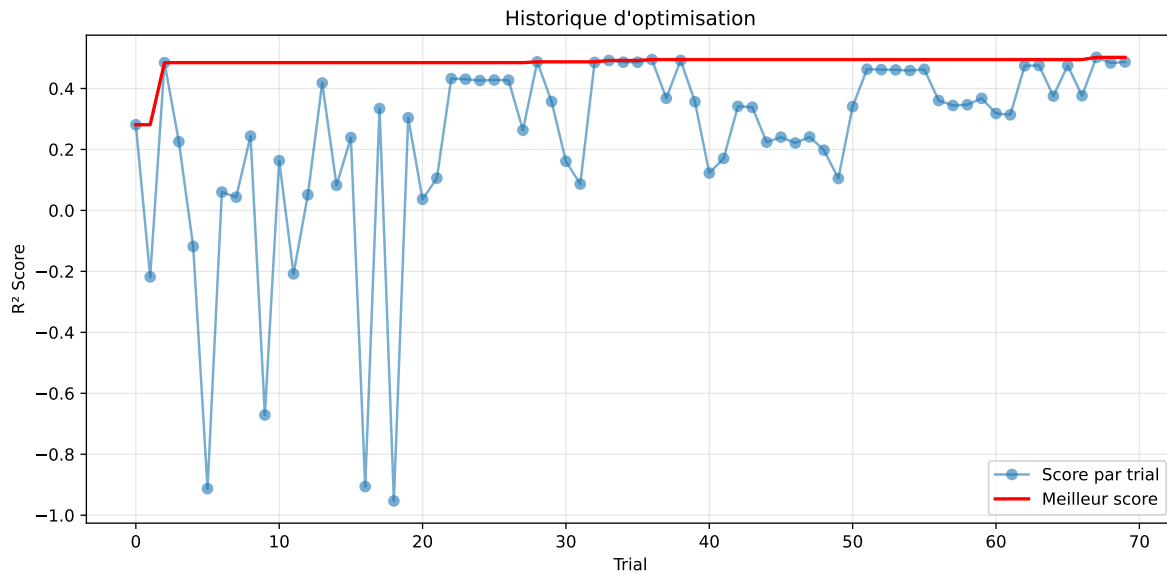


## 2.5 Model : CatBoost

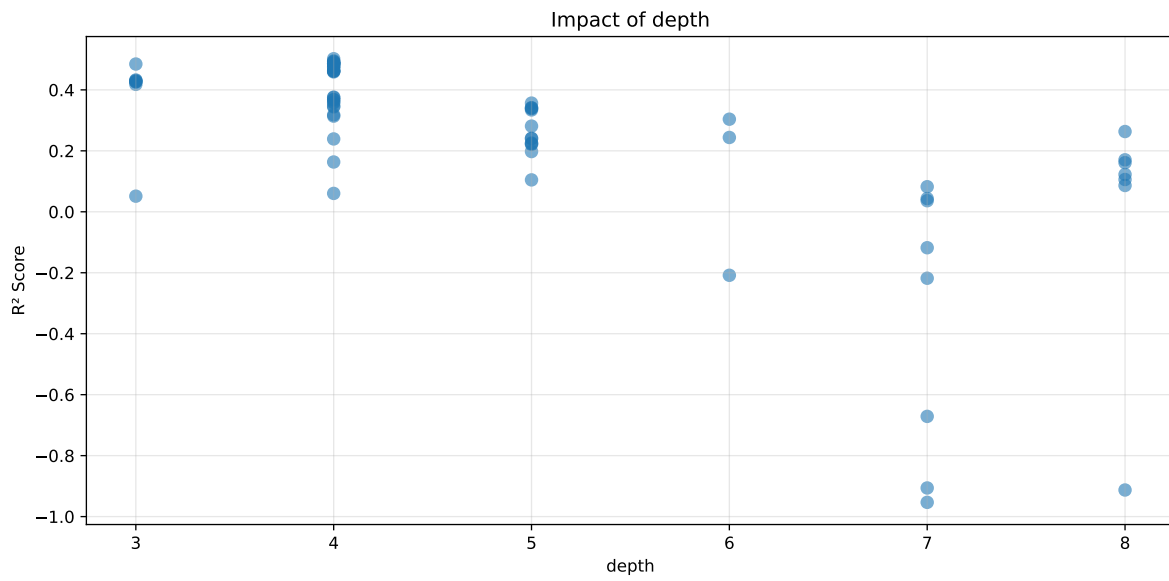
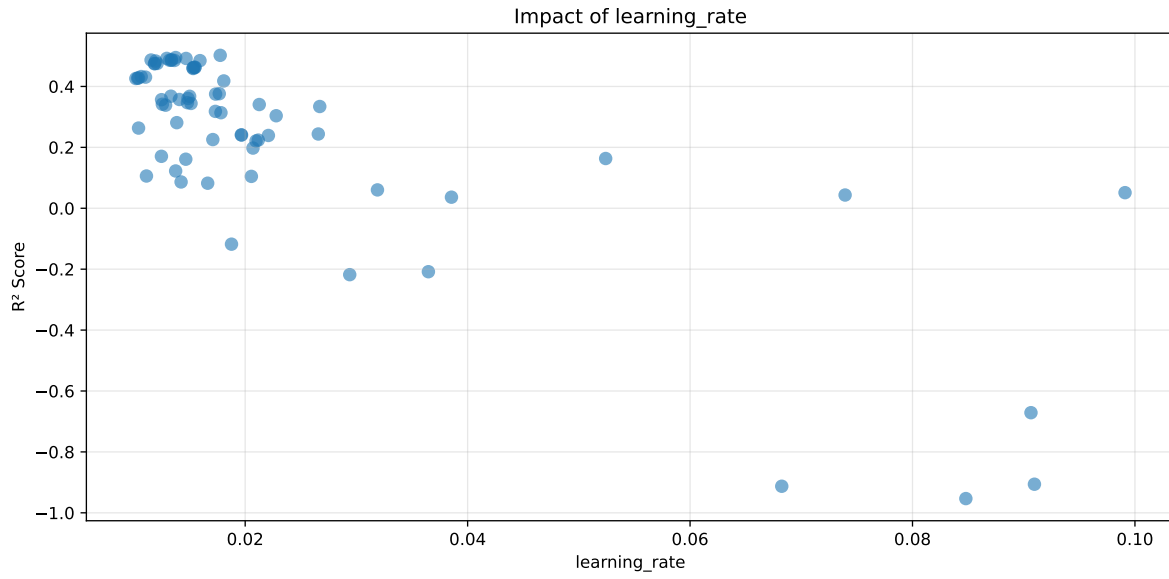
### 2.5.1 Gridsearch

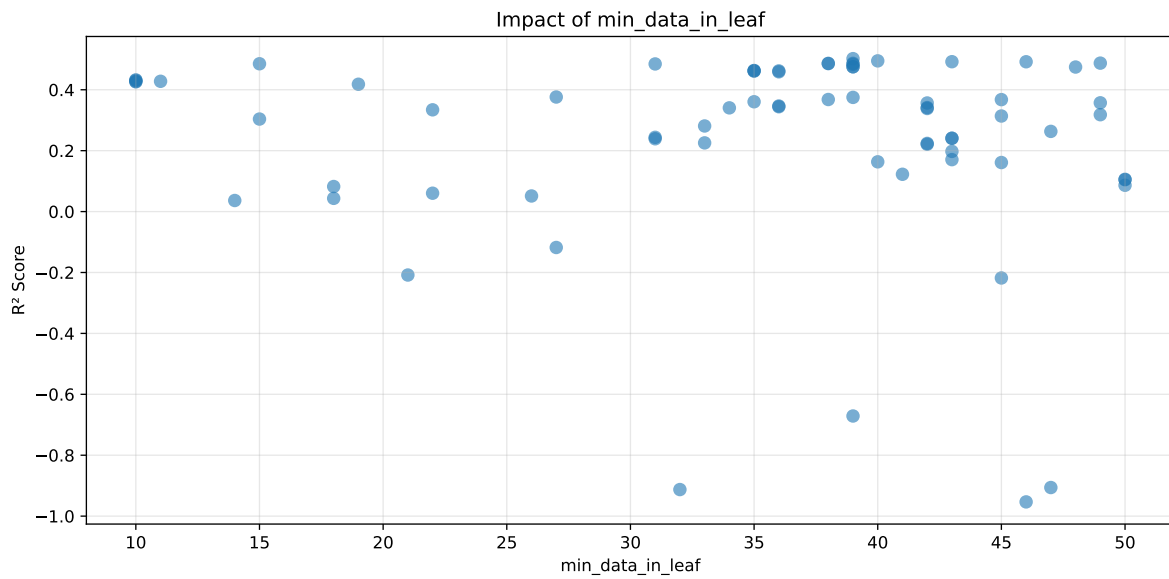
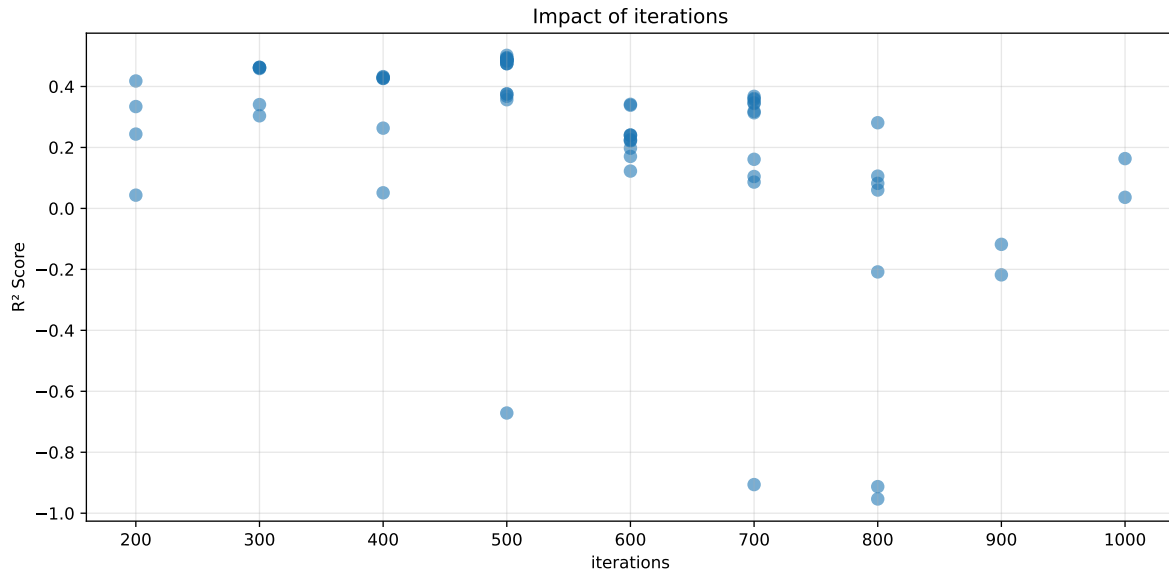
Distribution y\_train vs y\_test: y\_train: mean=2.339, std=0.798, min=0.959, max=13.845  
y\_test: mean=2.374, std=0.786, min=1.071, max=7.277

===== TOP Importance FEATURES =====  
feature importance r\_umidity 42.401798 r\_NH3 20.739341 r\_CO2 17.909259 r\_THI 9.962510  
r\_temperature 8.987091



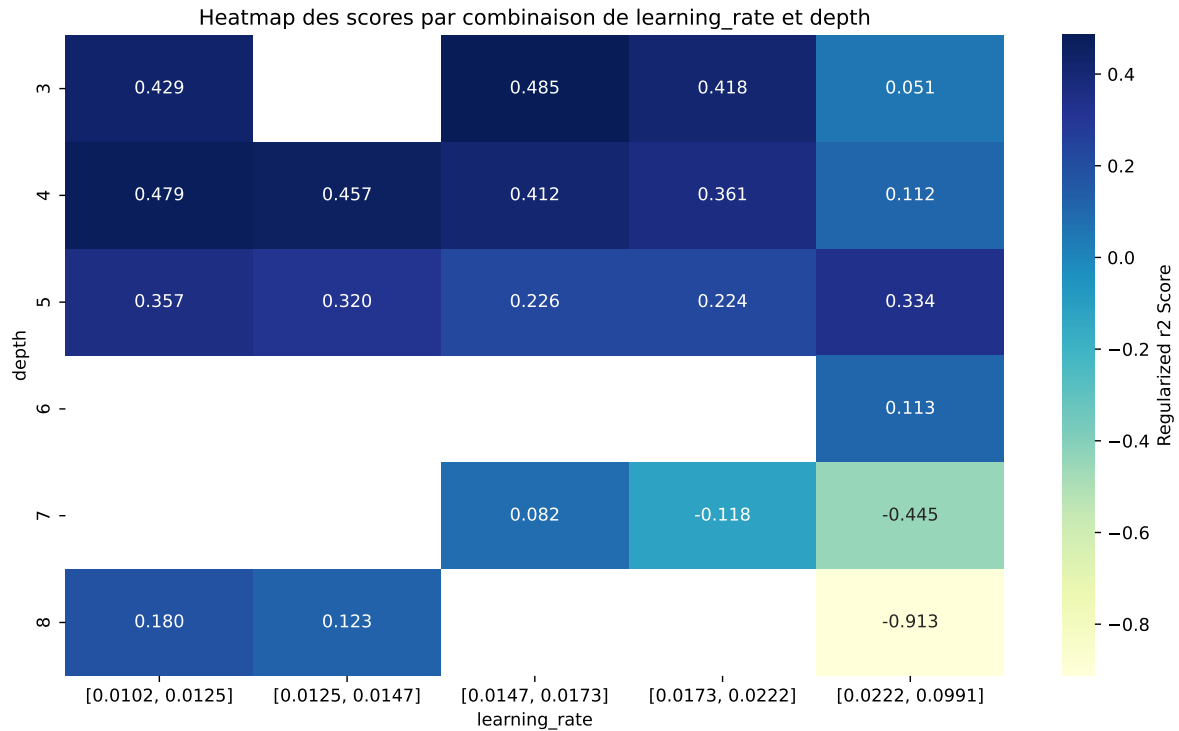
Paramètres avec >5% d'importance : ['learning\_rate', 'depth', 'iterations', 'min\_data\_in\_leaf']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine\_learning\workflow\_main

The default value of observed=False is deprecated and will change to observed=True in a future



| rank | mean_train_cv | mean_test_cv | test_score_cv | std_test_cv | penalized_score | test   | diff   | r_test | r_train |
|------|---------------|--------------|---------------|-------------|-----------------|--------|--------|--------|---------|
| 1    | 0.5264        | 0.5022       | 0.5216        | 0.0378      | 0.5022          | 0.5277 | 0.0256 | 1.0509 | 1.0482  |
| 2    | 0.5192        | 0.4949       | 0.5111        | 0.0373      | 0.4949          | 0.516  | 0.0211 | 1.0427 | 1.0491  |
| 3    | 0.5159        | 0.4921       | 0.5092        | 0.037       | 0.4921          | 0.5133 | 0.0211 | 1.0429 | 1.0482  |
| 4    | 0.5143        | 0.4919       | 0.5098        | 0.038       | 0.4919          | 0.516  | 0.0241 | 1.049  | 1.0454  |
| 5    | 0.5083        | 0.4874       | 0.5027        | 0.0365      | 0.4874          | 0.5086 | 0.0212 | 1.0434 | 1.0428  |
| 6    | 0.5077        | 0.4868       | 0.5027        | 0.0371      | 0.4868          | 0.5083 | 0.0215 | 1.0442 | 1.0429  |
| 7    | 0.5069        | 0.4864       | 0.5028        | 0.0381      | 0.4864          | 0.507  | 0.0206 | 1.0424 | 1.0422  |
| 8    | 0.5057        | 0.4862       | 0.5013        | 0.0377      | 0.4862          | 0.5081 | 0.0219 | 1.0451 | 1.0401  |
| 9    | 0.5062        | 0.4854       | 0.501         | 0.0378      | 0.4854          | 0.5084 | 0.023  | 1.0473 | 1.0428  |
| 10   | 0.5063        | 0.4848       | 0.4976        | 0.0349      | 0.4848          | 0.4998 | 0.015  | 1.031  | 1.0444  |
| ---  | ---           | ---          | ---           | ---         | ---             | ---    | ---    | ---    | ---     |
| 70   | 0.8784        | 0.5731       | 0.6112        | 0.0366      | -0.9533         | 0.6257 | 0.0526 | 1.0917 | 1.5327  |

### Hyperparameters:

Rank 1: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.017753764737520203, 'reg\_\_l2\_leaf\_reg': 16.113783981554867, 'reg\_\_bagging\_temperature': 2.6177764545288085, 'reg\_\_random\_strength': 4.462369896473606, 'reg\_\_subsample': 0.755081898525712, 'reg\_\_min\_data\_in\_leaf': 39, 'reg\_\_leaf\_estimation\_iterations': 10}



Rank 2: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.013748202799950676, 'reg\_\_l2\_leaf\_reg': 4.6598532849077205, 'reg\_\_bagging\_temperature': 2.3373887519280117, 'reg\_\_random\_strength': 3.1644922141889724, 'reg\_\_subsample': 0.7463923697431301, 'reg\_\_min\_data\_in\_leaf': 40, 'reg\_\_leaf\_estimation\_iterations': 9}

Rank 3: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.012950988960206583, 'reg\_\_l2\_leaf\_reg': 6.5715612647424075, 'reg\_\_bagging\_temperature': 2.344463727474487, 'reg\_\_random\_strength': 2.9607114726851576, 'reg\_\_subsample': 0.8052642385057212, 'reg\_\_min\_data\_in\_leaf': 43, 'reg\_\_leaf\_estimation\_iterations': 9}

Rank 4: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.014683986353605884, 'reg\_\_l2\_leaf\_reg': 8.652733351195291, 'reg\_\_bagging\_temperature': 2.4519739327961303, 'reg\_\_random\_strength': 4.025396722129051, 'reg\_\_subsample': 0.7876903036951233, 'reg\_\_min\_data\_in\_leaf': 46, 'reg\_\_leaf\_estimation\_iterations': 9}

Rank 5: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.013383933036618226, 'reg\_\_l2\_leaf\_reg': 1.3957799070999148, 'reg\_\_bagging\_temperature': 2.360587471608662, 'reg\_\_random\_strength': 4.108320174924199, 'reg\_\_subsample': 0.7773944266142561, 'reg\_\_min\_data\_in\_leaf': 49, 'reg\_\_leaf\_estimation\_iterations': 10}

Rank 6: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.01153800973158288, 'reg\_\_l2\_leaf\_reg': 16.609871002673046, 'reg\_\_bagging\_temperature': 2.597844561024238, 'reg\_\_random\_strength': 2.6324359437818834, 'reg\_\_subsample': 0.7534723683861687, 'reg\_\_min\_data\_in\_leaf': 39, 'reg\_\_leaf\_estimation\_iterations': 10}

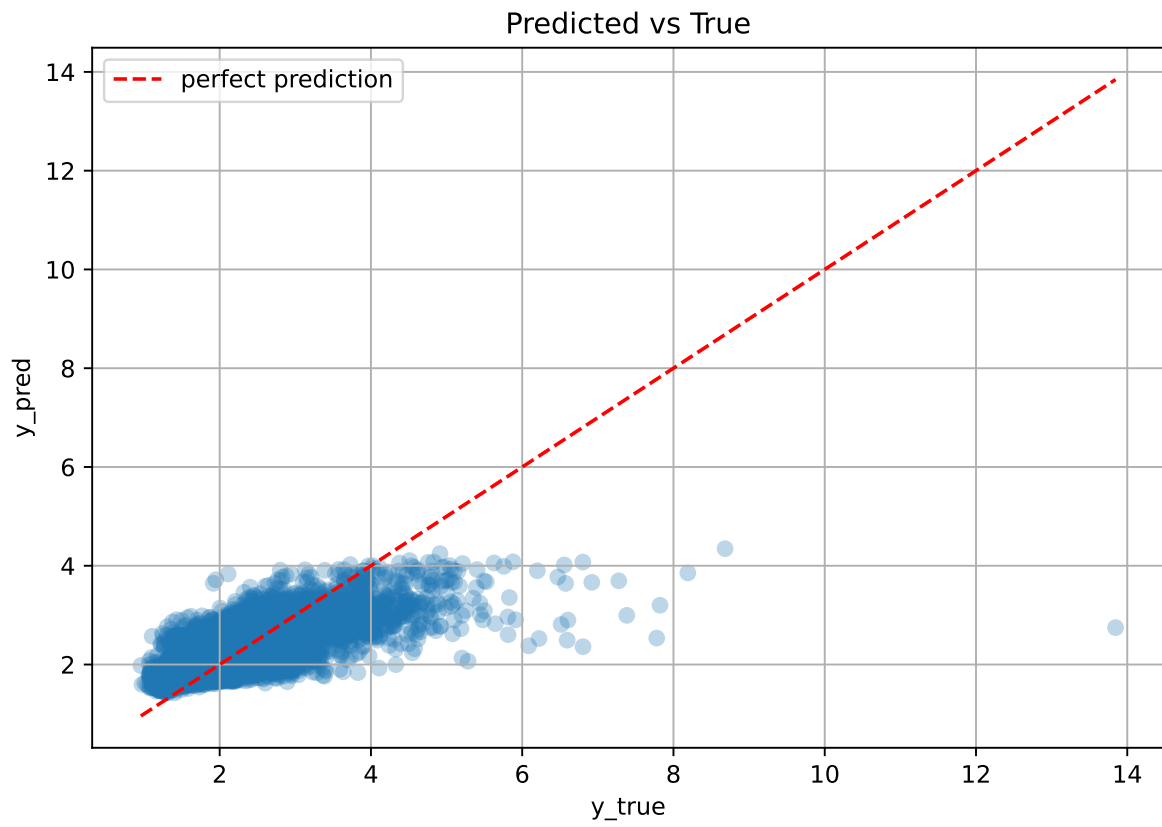
Rank 7: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.013359958338071678, 'reg\_\_l2\_leaf\_reg': 8.056055328978427, 'reg\_\_bagging\_temperature': 2.307268258233183, 'reg\_\_random\_strength': 4.099732073891053, 'reg\_\_subsample': 0.7907868546335329, 'reg\_\_min\_data\_in\_leaf': 38, 'reg\_\_leaf\_estimation\_iterations': 9}

Rank 8: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.01319671273136439, 'reg\_\_l2\_leaf\_reg': 8.09798059876641, 'reg\_\_bagging\_temperature': 2.4206766239540904, 'reg\_\_random\_strength': 4.039976744508566, 'reg\_\_subsample': 0.7508081720327328, 'reg\_\_min\_data\_in\_leaf': 38, 'reg\_\_leaf\_estimation\_iterations': 9}

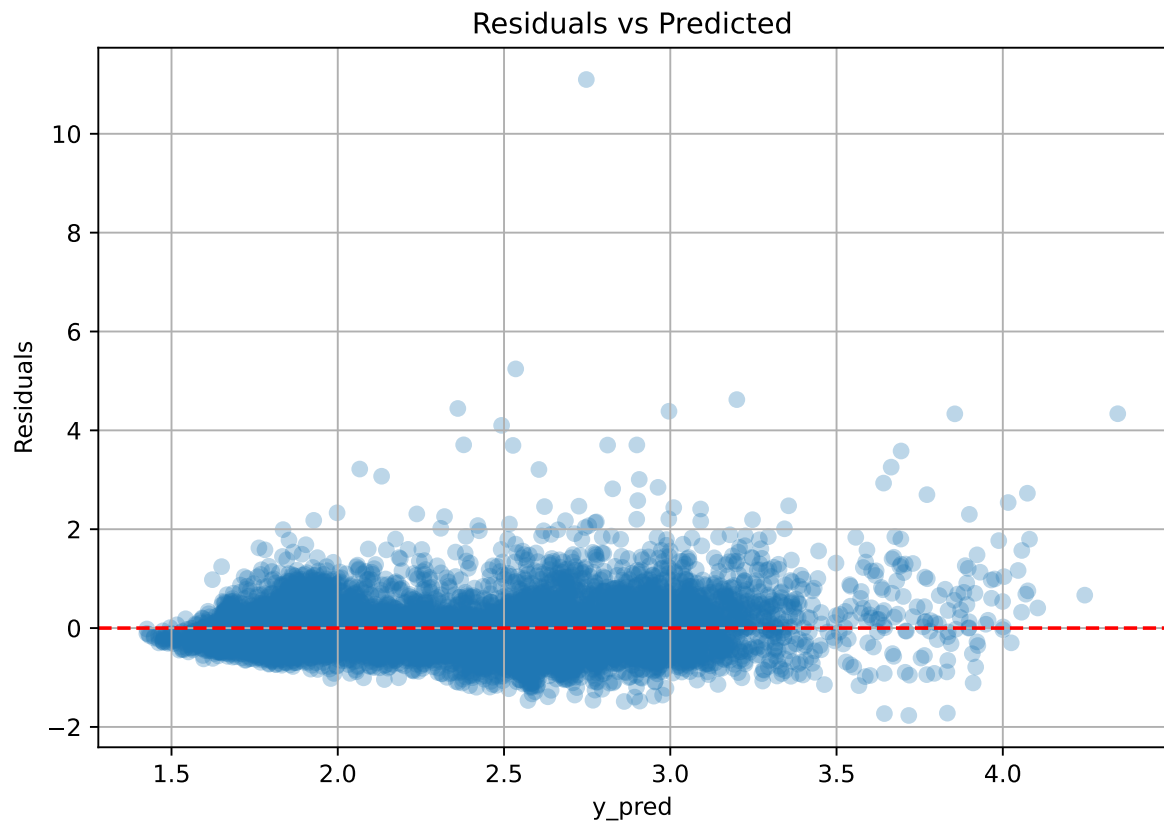
Rank 9: {'reg\_\_iterations': 500, 'reg\_\_depth': 4, 'reg\_\_learning\_rate': 0.013642245780249572, 'reg\_\_l2\_leaf\_reg': 7.6543818429520325, 'reg\_\_bagging\_temperature': 2.4326327869261846, 'reg\_\_random\_strength': 4.210003538965745, 'reg\_\_subsample': 0.6718213890224929, 'reg\_\_min\_data\_in\_leaf': 15, 'reg\_\_leaf\_estimation\_iterations': 8}

Rank 10: {'reg\_\_iterations': 500, 'reg\_\_depth': 3, 'reg\_\_learning\_rate': 0.015936057632381605, 'reg\_\_l2\_leaf\_reg': 3.937915783665093, 'reg\_\_bagging\_temperature': 2.916558242595164, 'reg\_\_random\_strength': 2.5227068969258037, 'reg\_\_subsample': 0.7343096391651582, 'reg\_\_min\_data\_in\_leaf': 31, 'reg\_\_leaf\_estimation\_iterations': 10}

### 2.5.2 Scatter



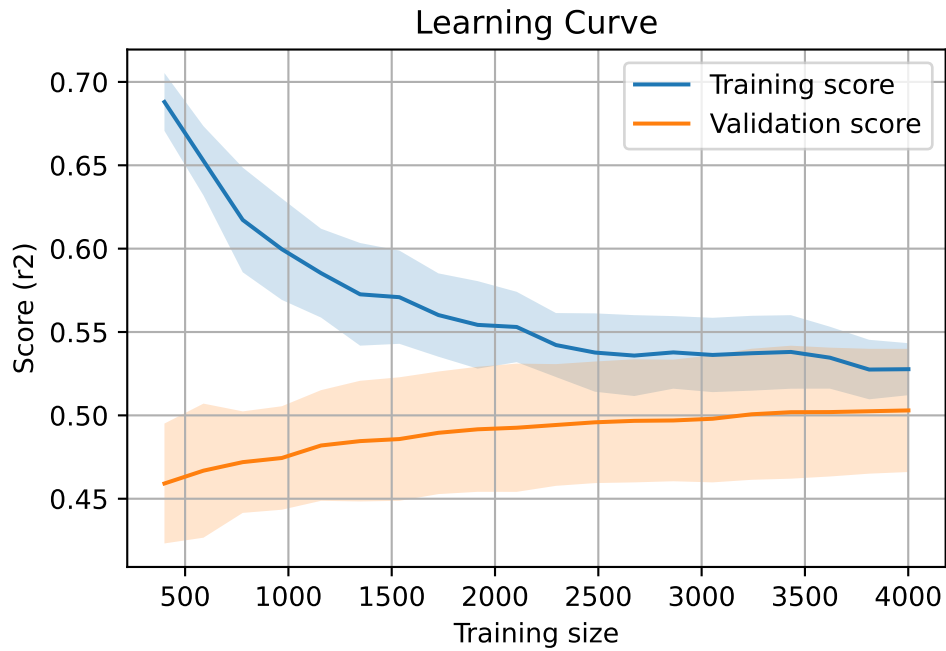
### 2.5.3 Scatter residuals



### 2.5.4 Learning curve

**Len(group)** : 8168, **Len(training)** : 5718,

**Len(training\_real)** : 4003, **Len(test\_of\_training)** : 1715, **Len(test)** : 2450



## 2.6 Sumup-table

| model    | mean_train_cv | mean_test_cv | std_test_cv | test_cv | test  | diff  | r_test | r_train | pen_score |
|----------|---------------|--------------|-------------|---------|-------|-------|--------|---------|-----------|
| Linear   | 0.36          | 0.365        | 0.037       | 0.345   | 0.345 | -0.02 | 0.946  | 0.986   | nan       |
| LGBM     | 0.469         | 0.448        | 0.037       | 0.46    | 0.475 | 0.027 | 1.06   | 1.047   | 0.4485    |
| CatBoost | 0.526         | 0.502        | 0.038       | 0.522   | 0.528 | 0.026 | 1.051  | 1.048   | 0.5022    |

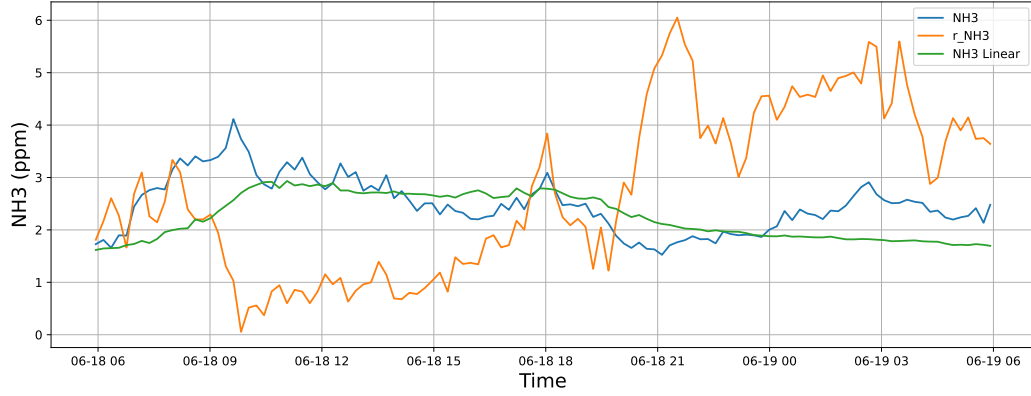
## 3 Plot

### 3.1 Standalone plots

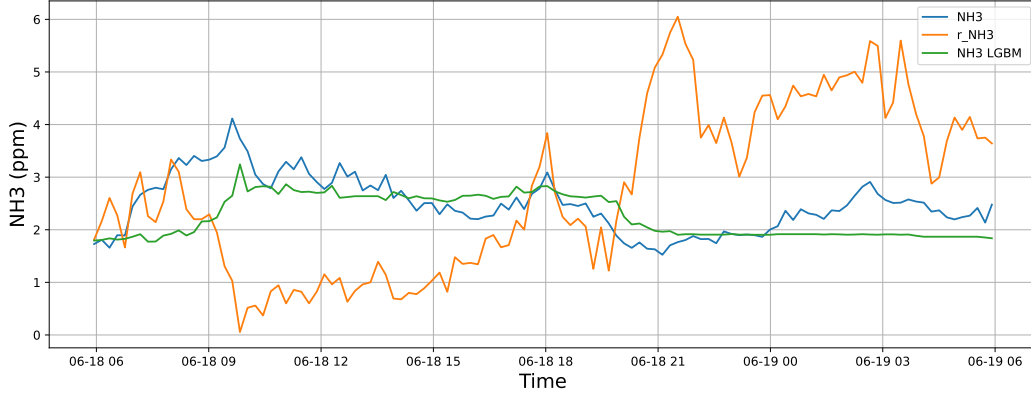
3.1.1 Group: campaign: 2025\_Jun,id\_rabbit: 1,id\_channel: 3

3.1.1.1 Time window : 24

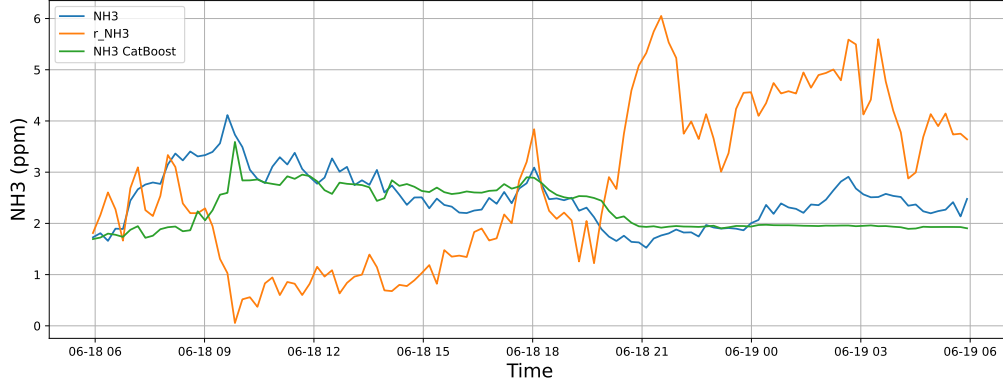
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-18 → 2025-06-19

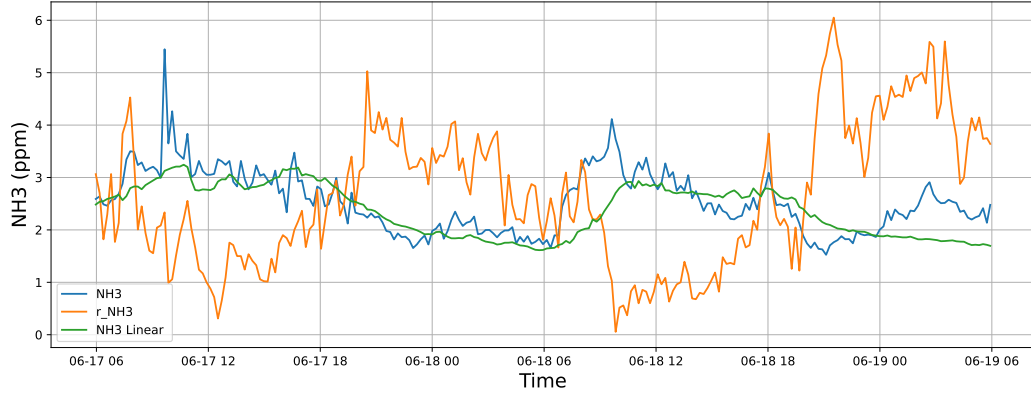


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-18 → 2025-06-19

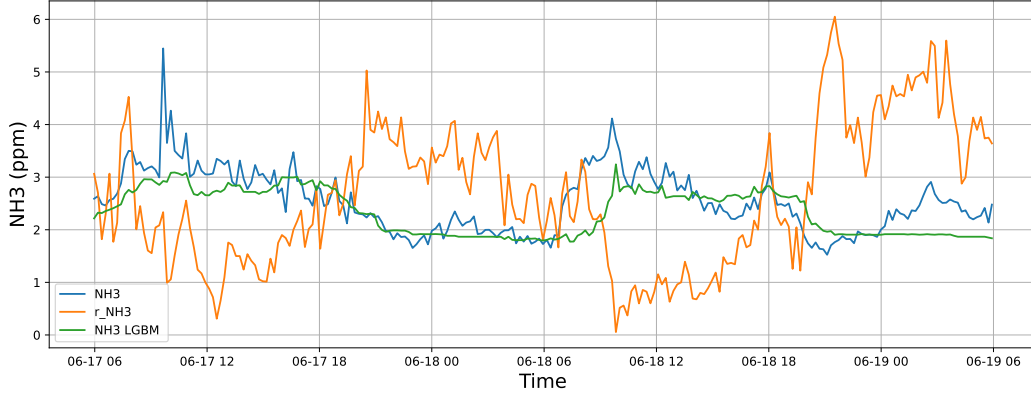


### 3.1.1.2 Time window : 48

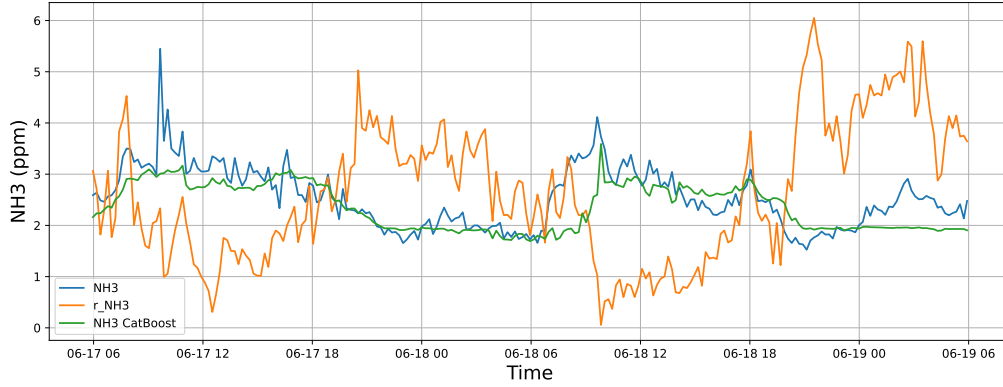
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-17 → 2025-06-19

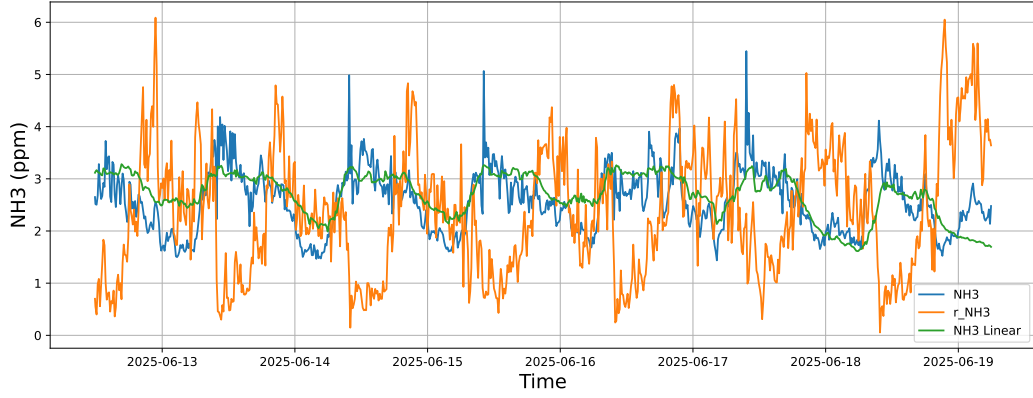


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-17 → 2025-06-19

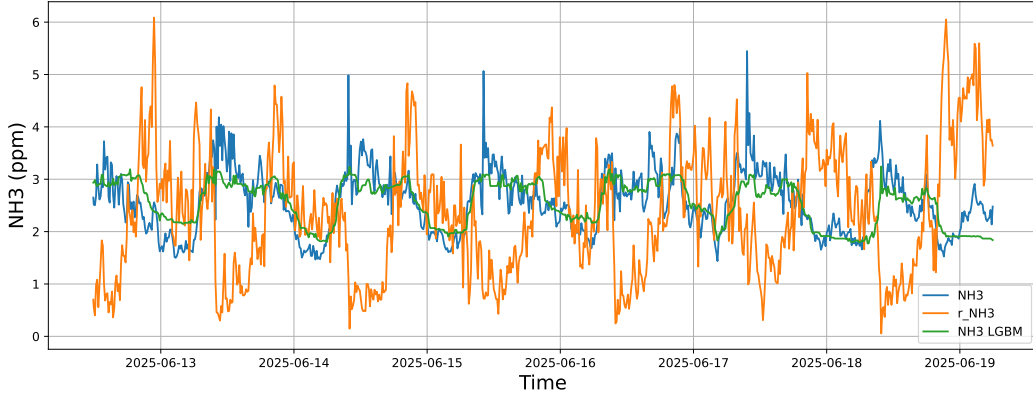


### 3.1.1.3 Time window : ALL

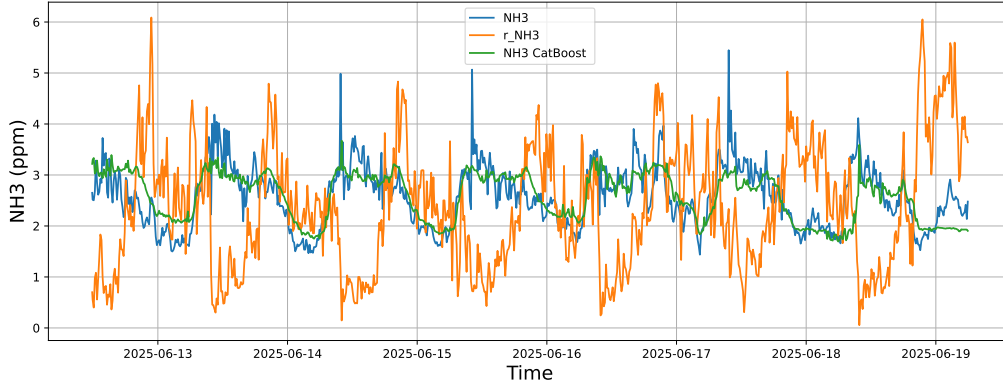
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-12 → 2025-06-19



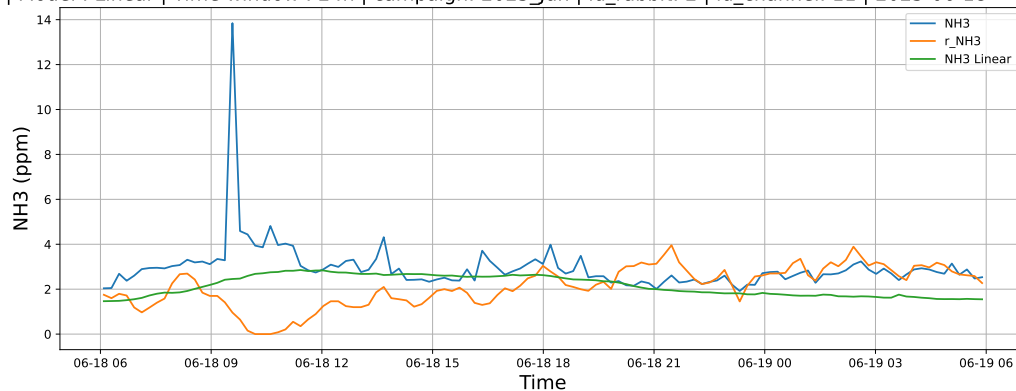
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-12 → 2025-06-19



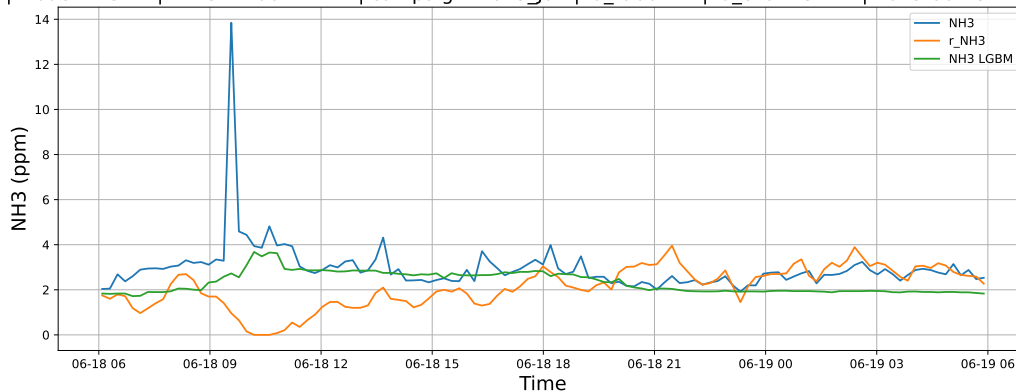
### 3.1.2 Group: campaign: 2025\_Jun,id\_rabbit: 2,id\_channel: 11

#### 3.1.2.1 Time window : 24

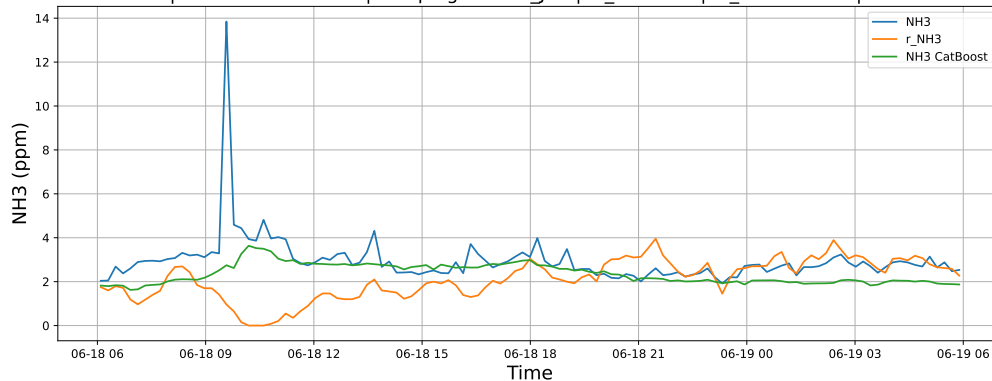
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-18 → 2025-06-19



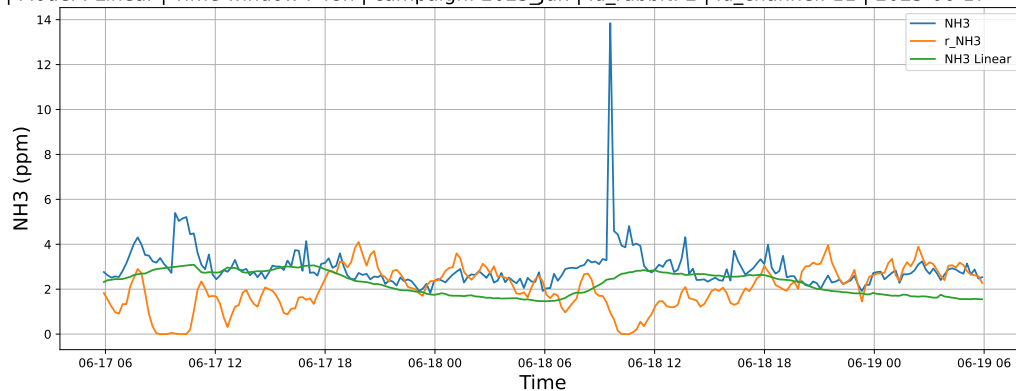
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-18 → 2025-06-19



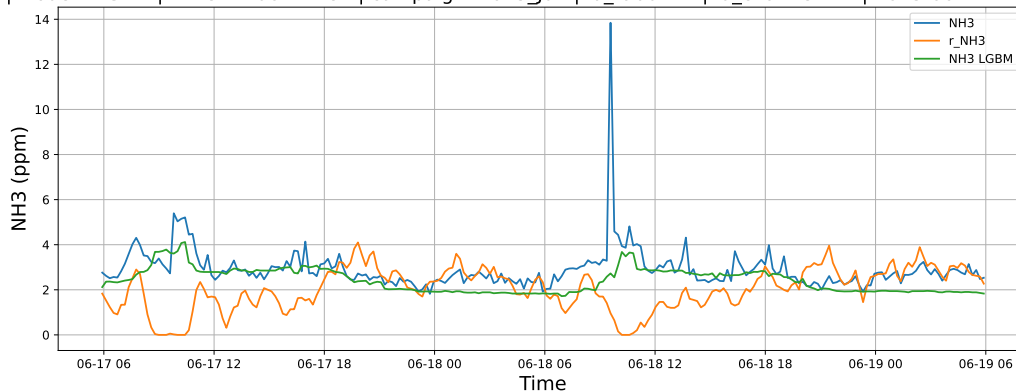
### 3.1.2.2 Time window : 48



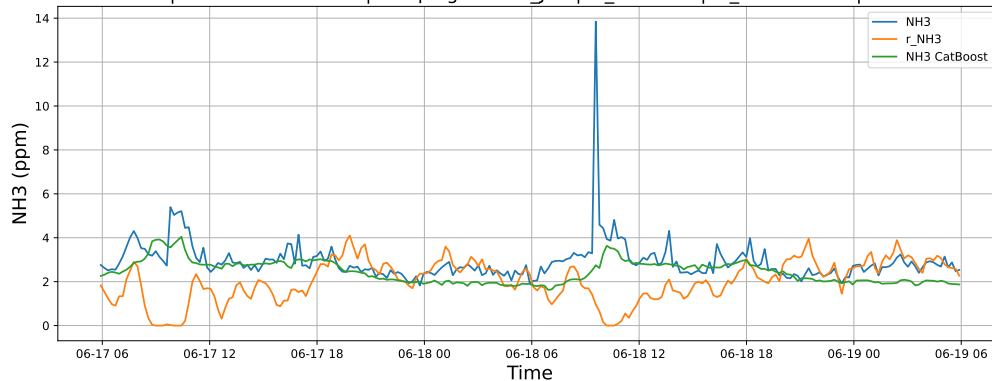
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-17 → 2025-06-19

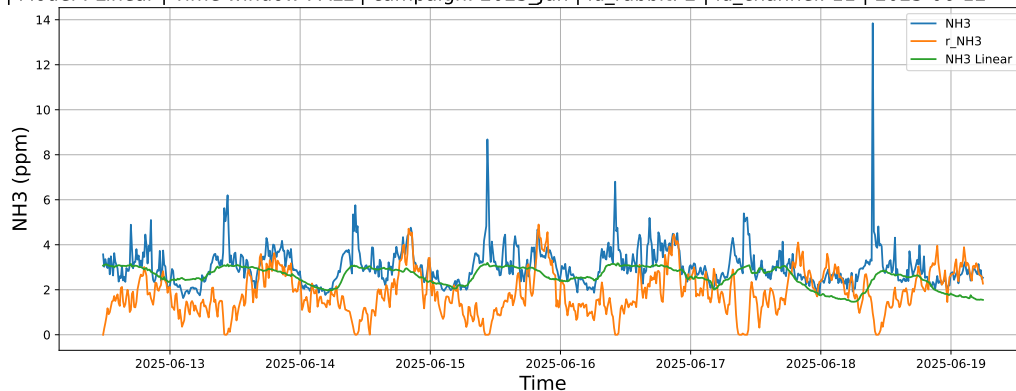


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-17 → 2025-06-19

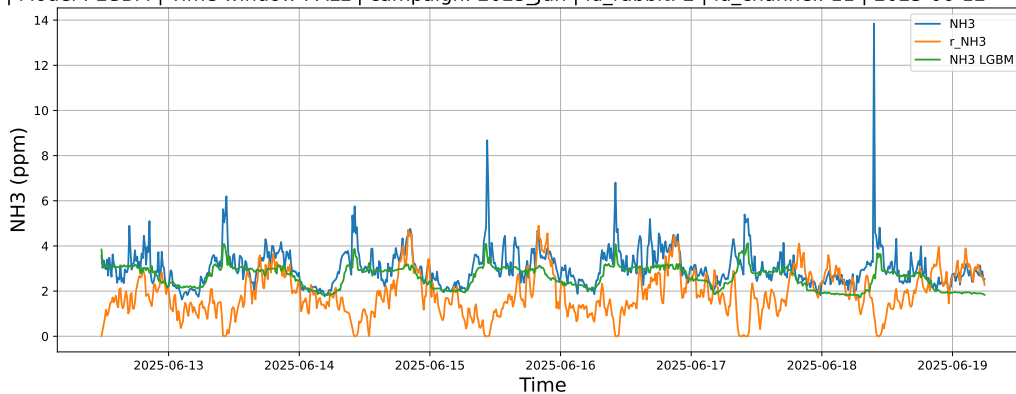


### 3.1.2.3 Time window : ALL

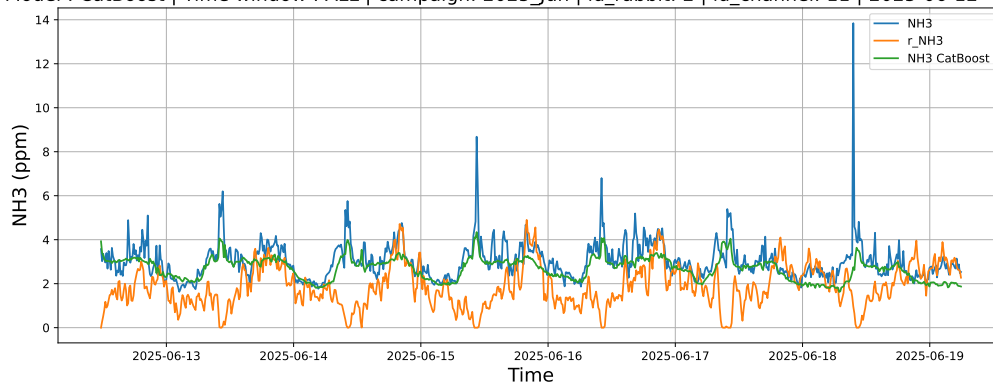
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-12 → 2025-06-19



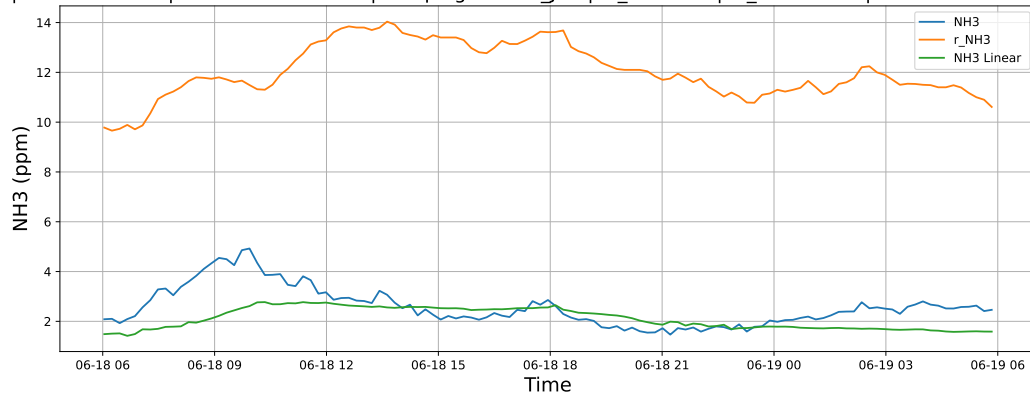
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-12 → 2025-06-19



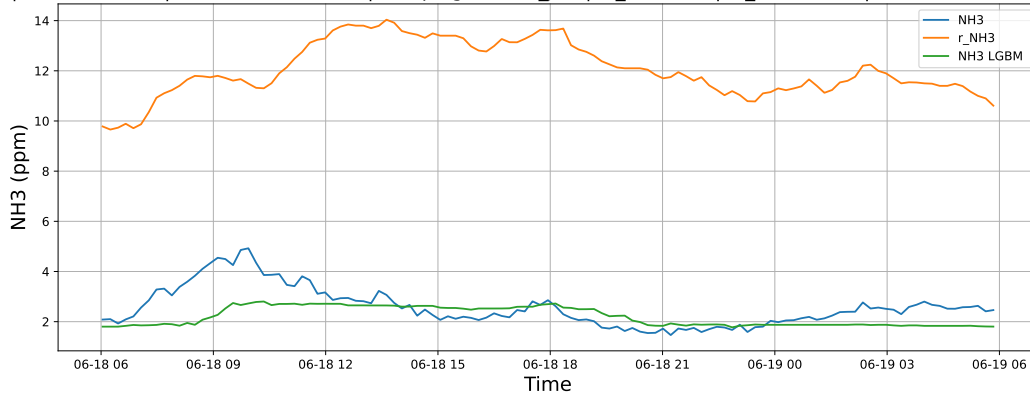
### 3.1.3 Group: campaign: 2025\_Jun,id\_rabbit: 4,id\_channel: 5

#### 3.1.3.1 Time window : 24

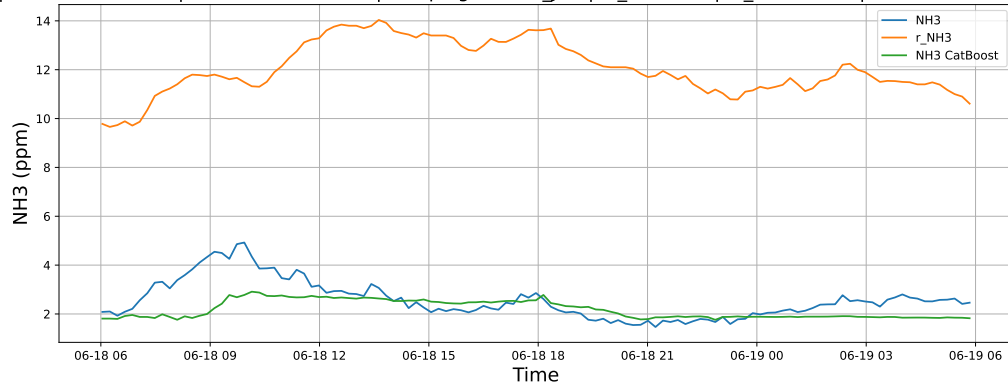
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-18 → 2025-06-19

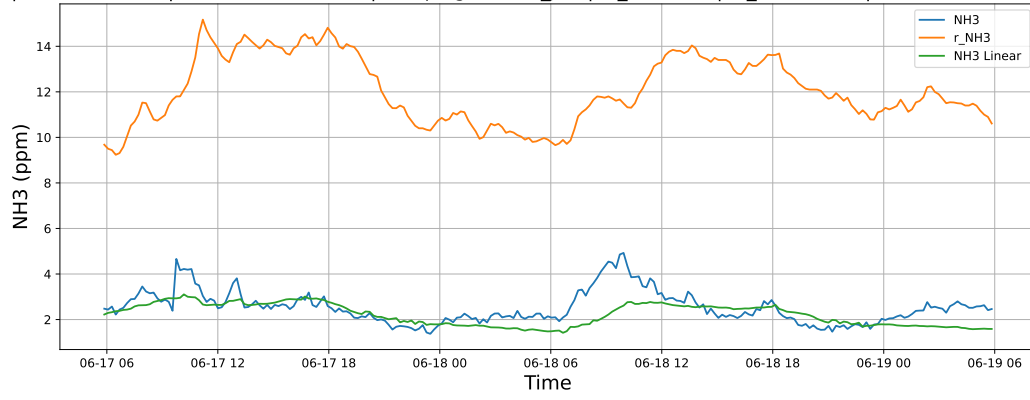


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-18 → 2025-06-19

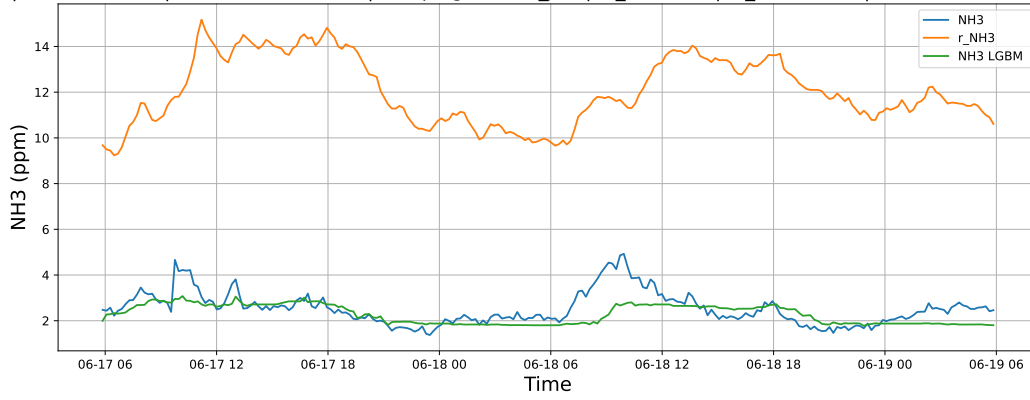


### 3.1.3.2 Time window : 48

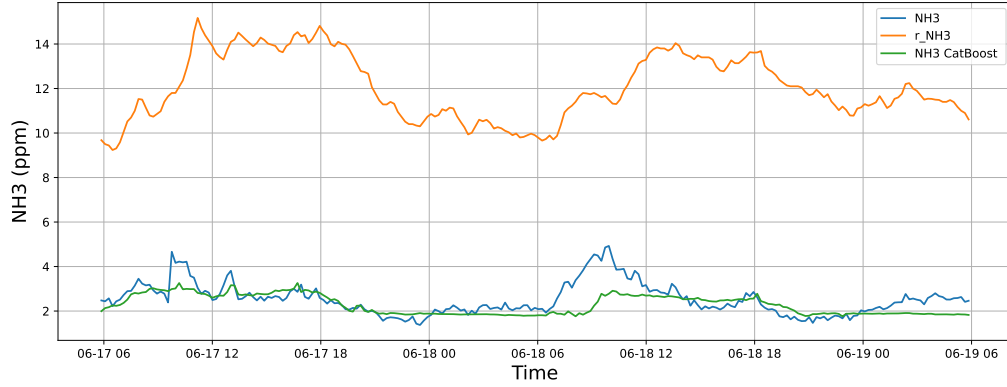
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-17 → 2025-06-19

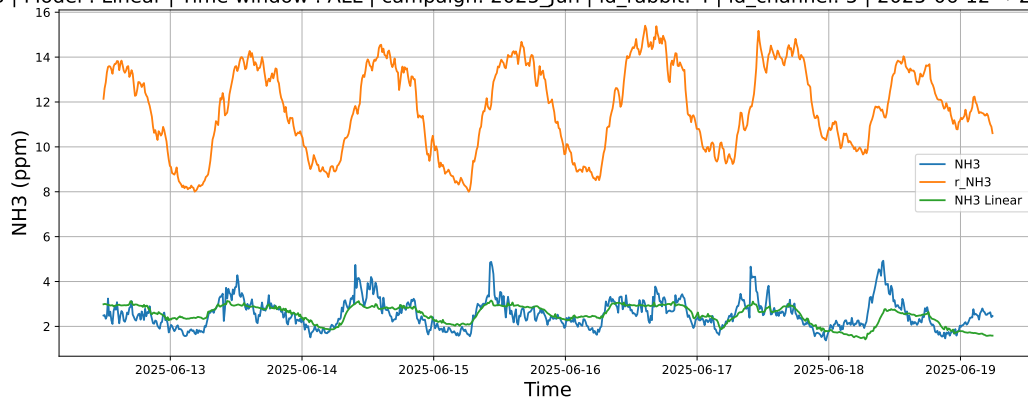


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-17 → 2025-06-19

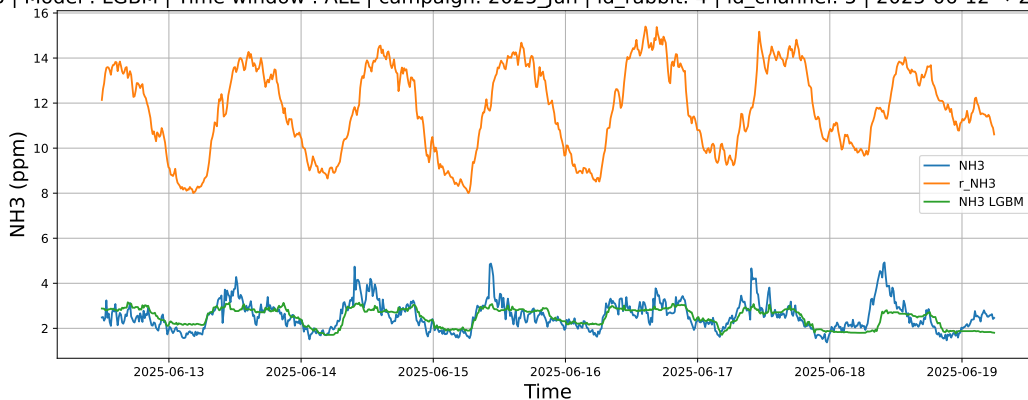


### 3.1.3.3 Time window : ALL

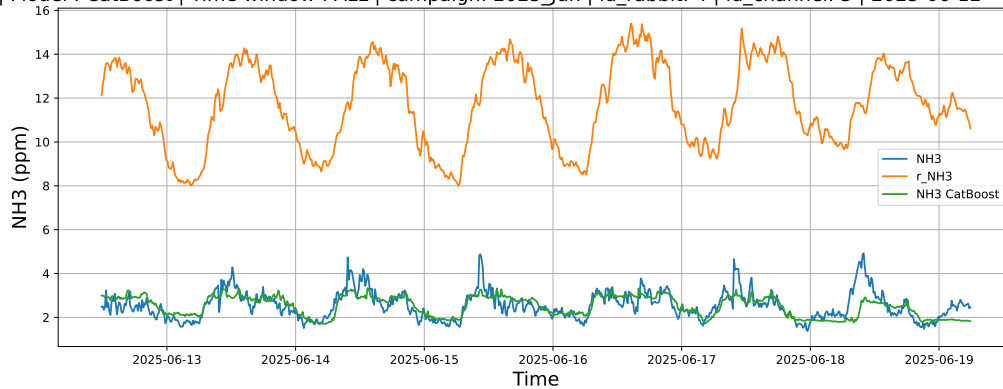
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-12 → 2025-06-19



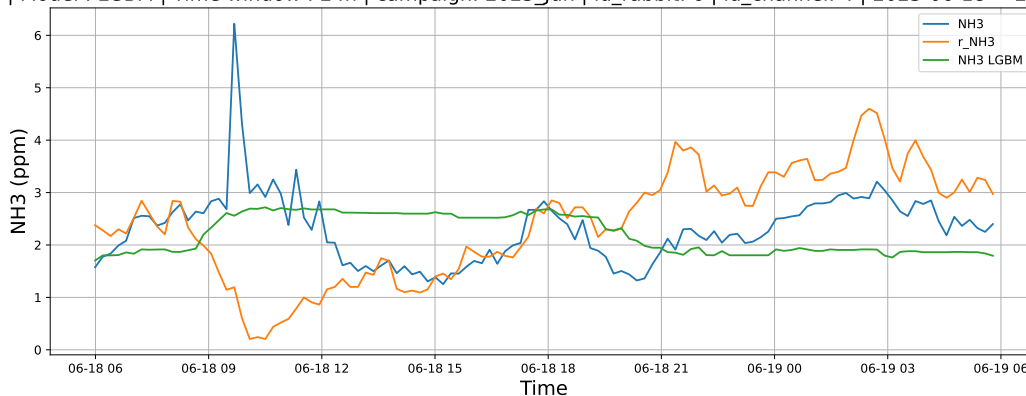
### 3.1.4 Group: campaign: 2025\_Jun,id\_rabbit: 6,id\_channel: 4

#### 3.1.4.1 Time window : 24

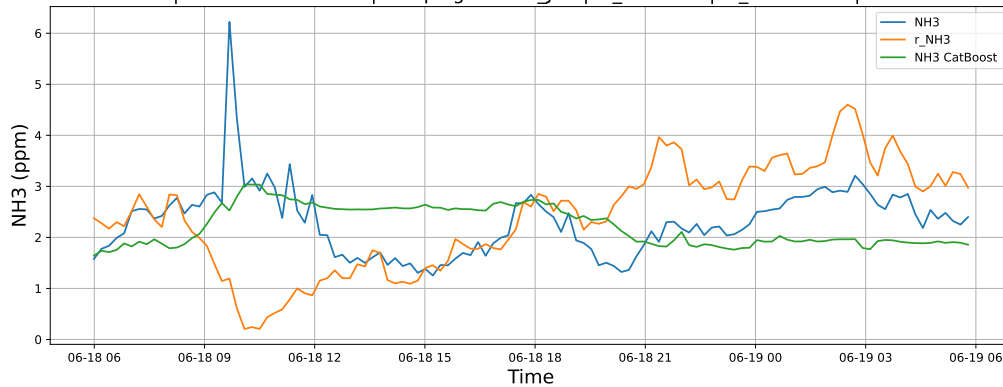
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-18 → 2025-06-19

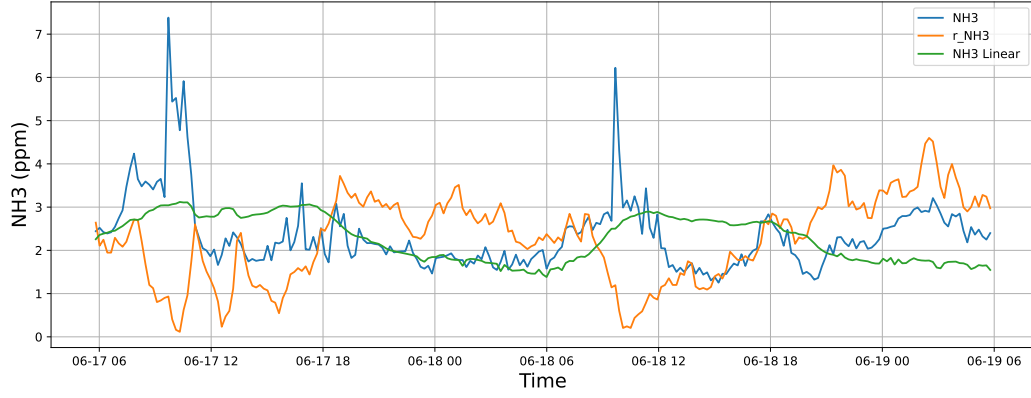


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-18 → 2025-06-19

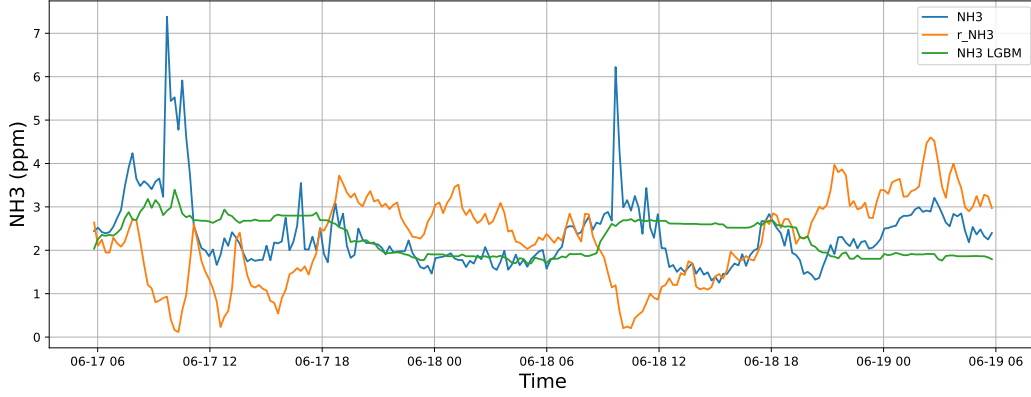


### 3.1.4.2 Time window : 48

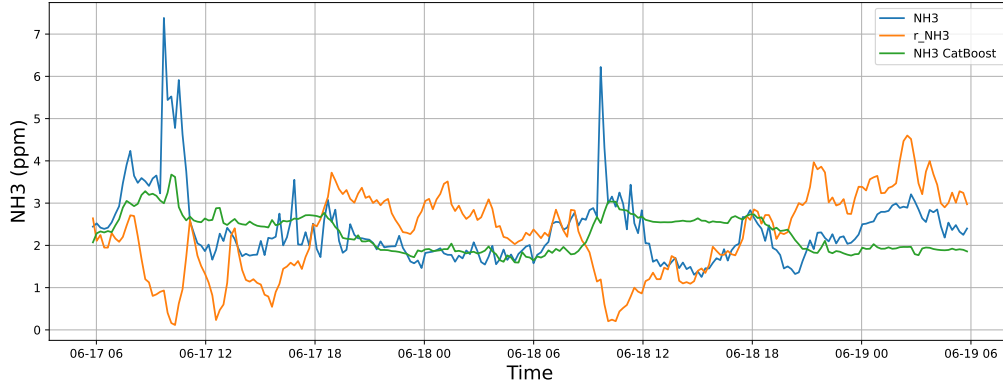
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-17 → 2025-06-19

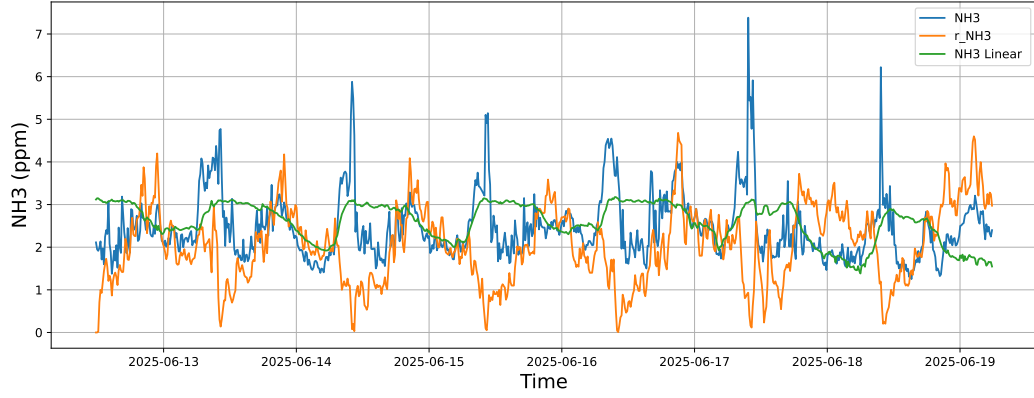


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-17 → 2025-06-19

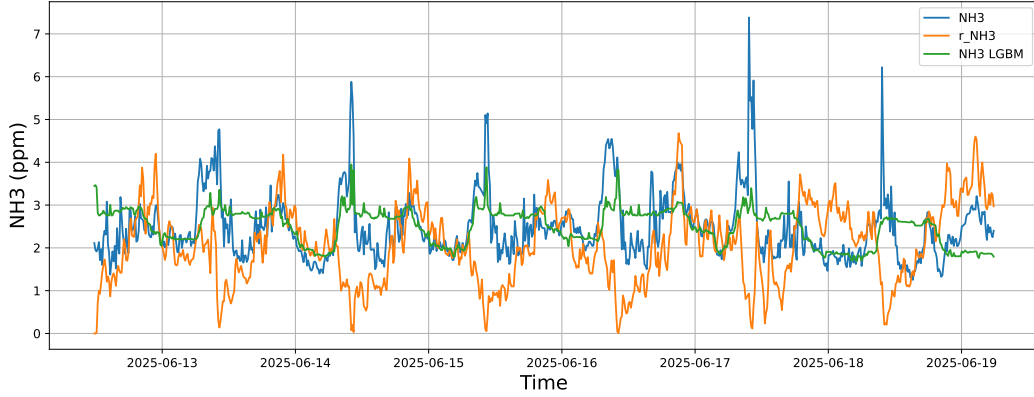


### 3.1.4.3 Time window : ALL

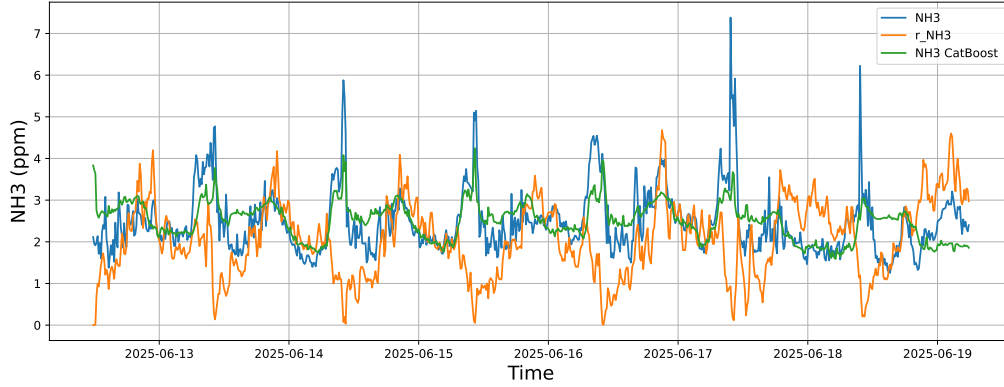
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-12 → 2025-06-19

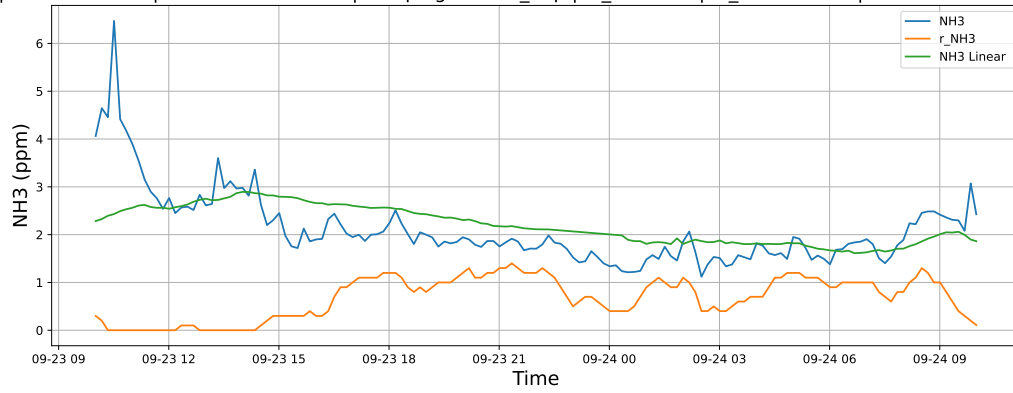


### 3.1.5 Group: campaign: 2025\_Sep,id\_rabbit: 2,id\_channel: 11

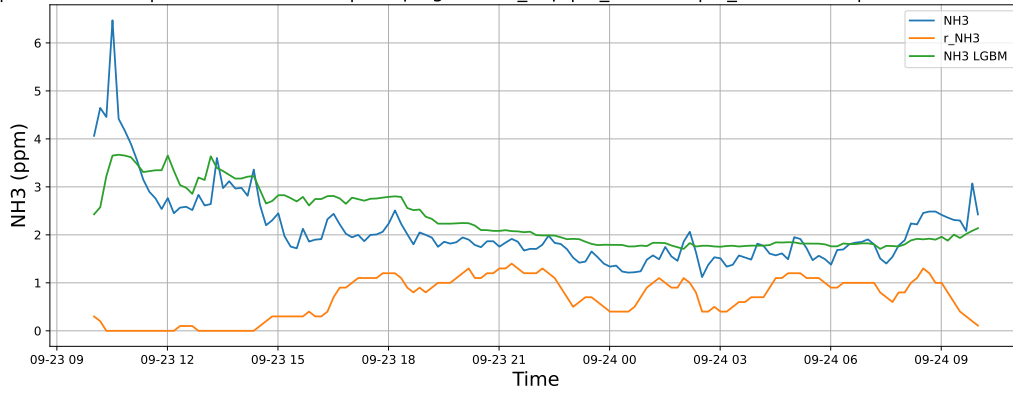
#### 3.1.5.1 Time window : 24



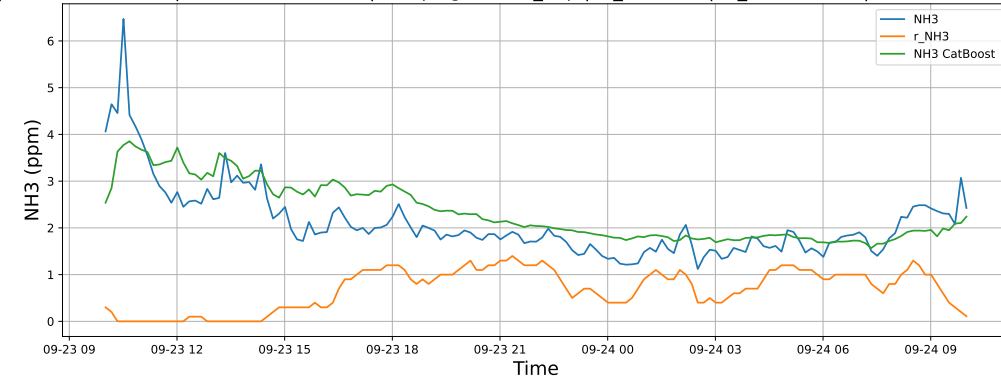
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-23 → 2025-09-24

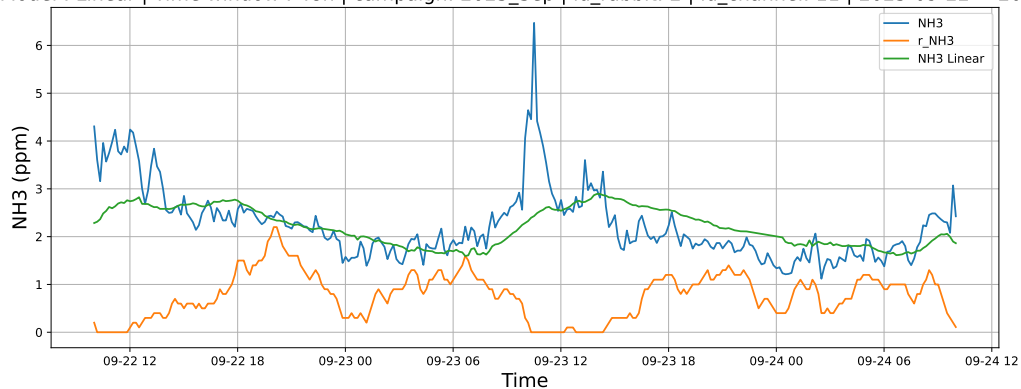


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-23 → 2025-09-24

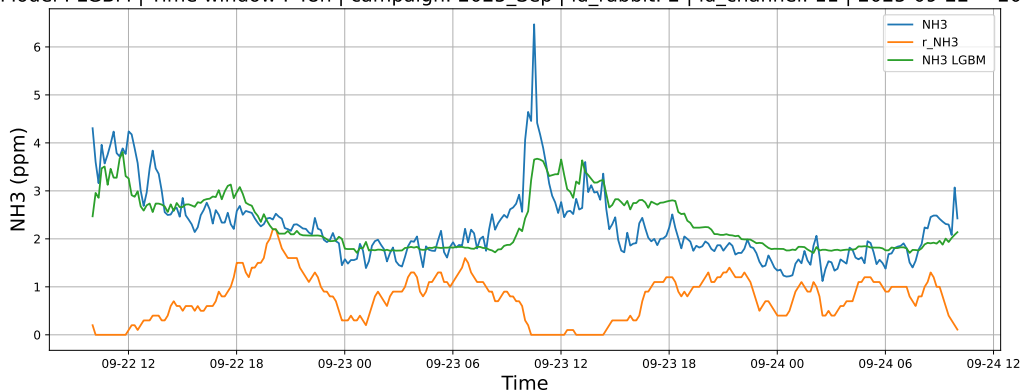


### 3.1.5.2 Time window : 48

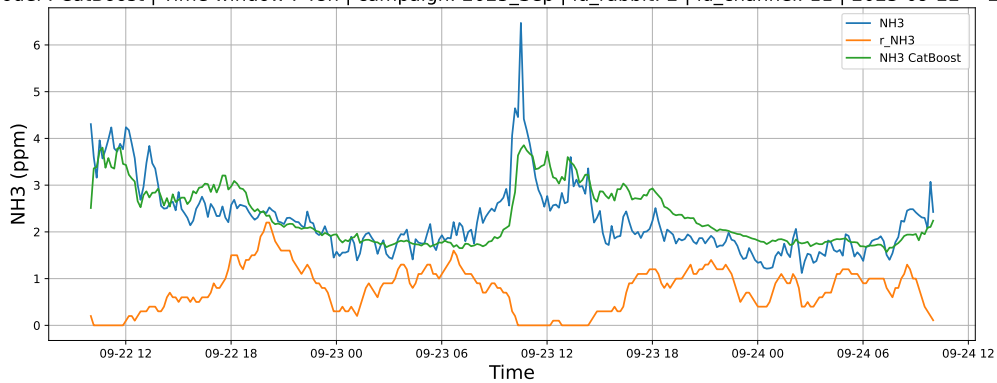
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-22 → 2025-09-24

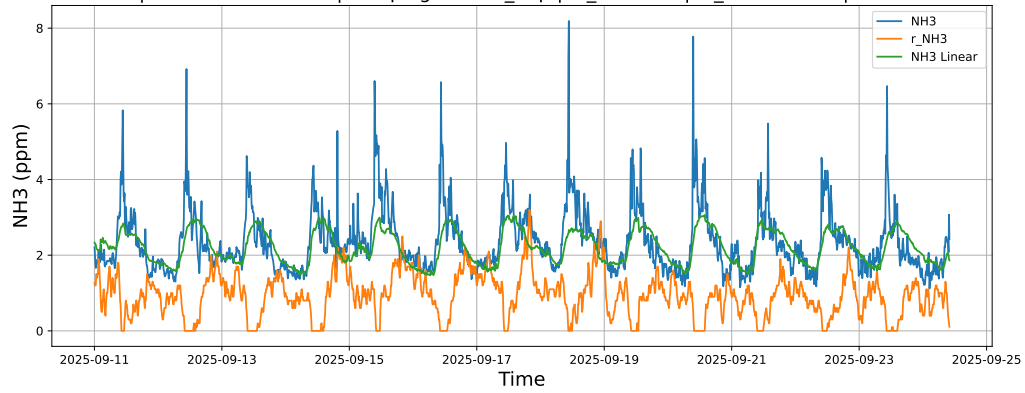


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-22 → 2025-09-24

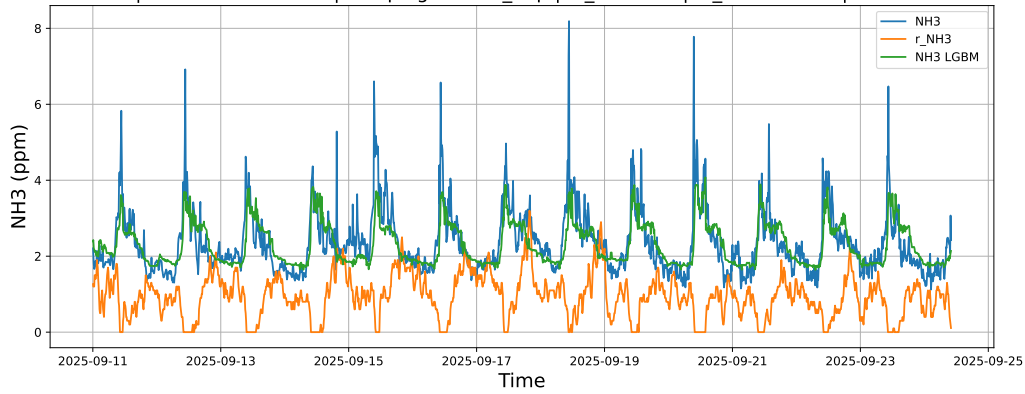


### 3.1.5.3 Time window : ALL

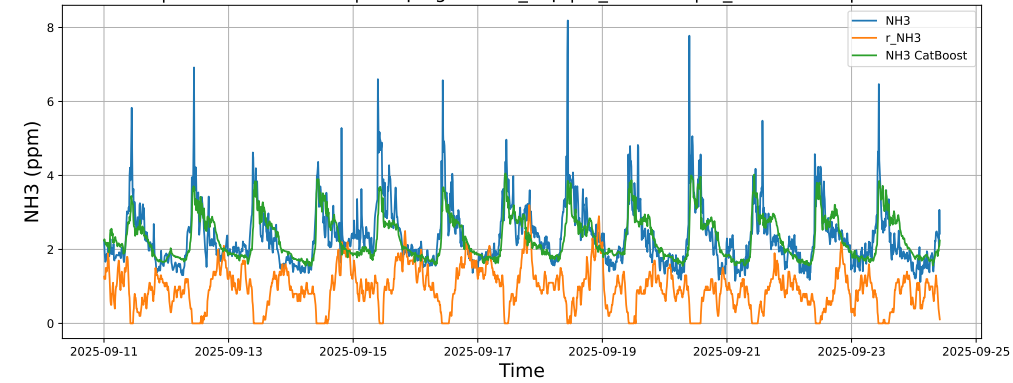
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-11 → 2025-09-24



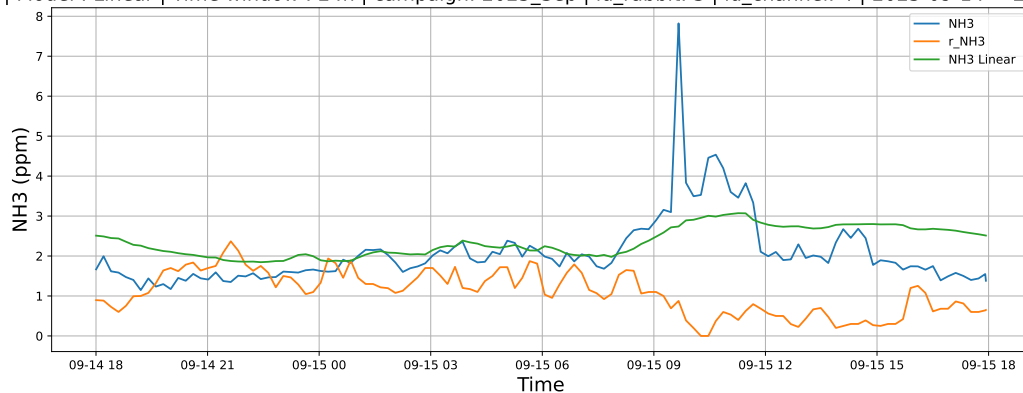
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-11 → 2025-09-24



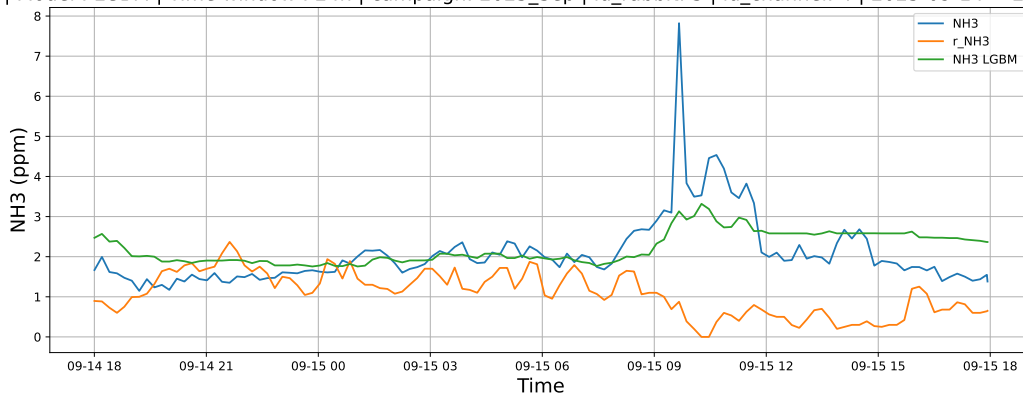
### 3.1.6 Group: campaign: 2025\_Sep,id\_rabbit: 3,id\_channel: 4

#### 3.1.6.1 Time window : 24

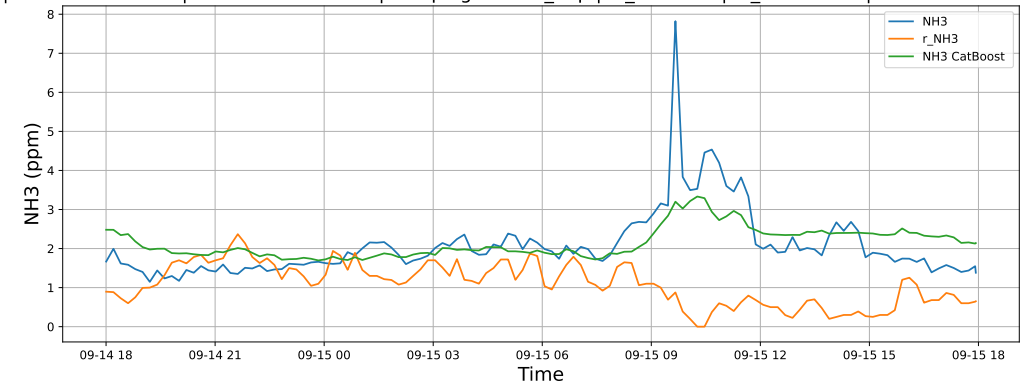
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-14 → 2025-09-15

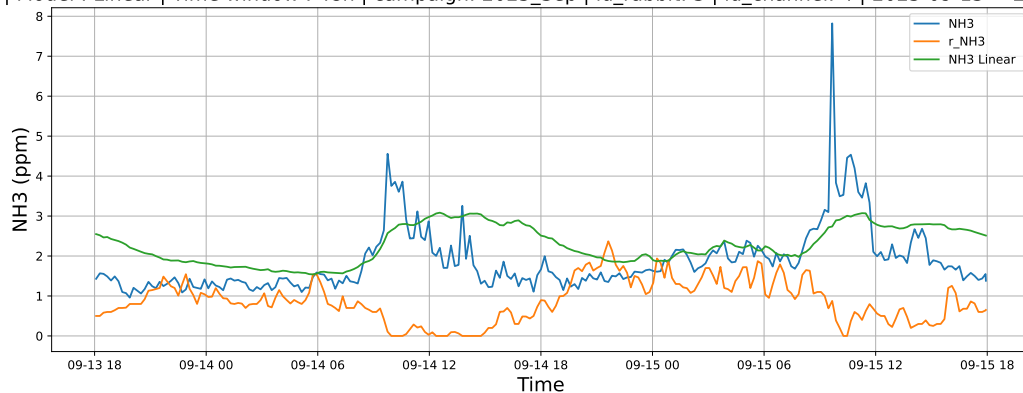


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-14 → 2025-09-15

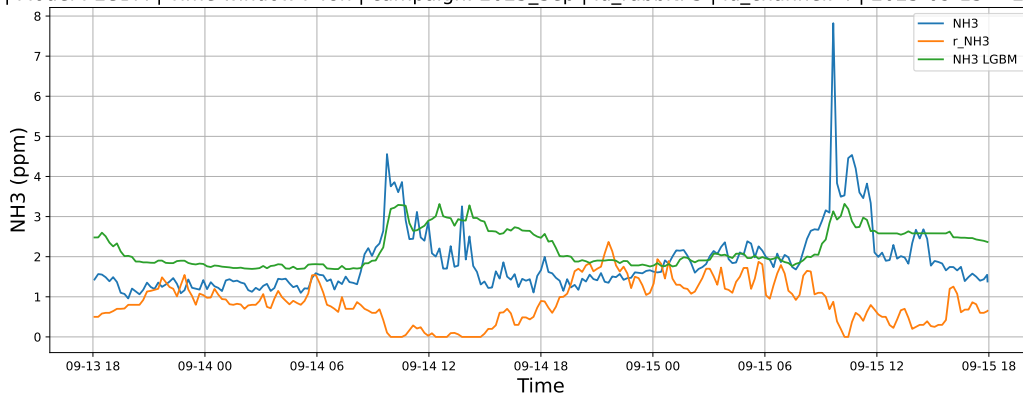


### 3.1.6.2 Time window : 48

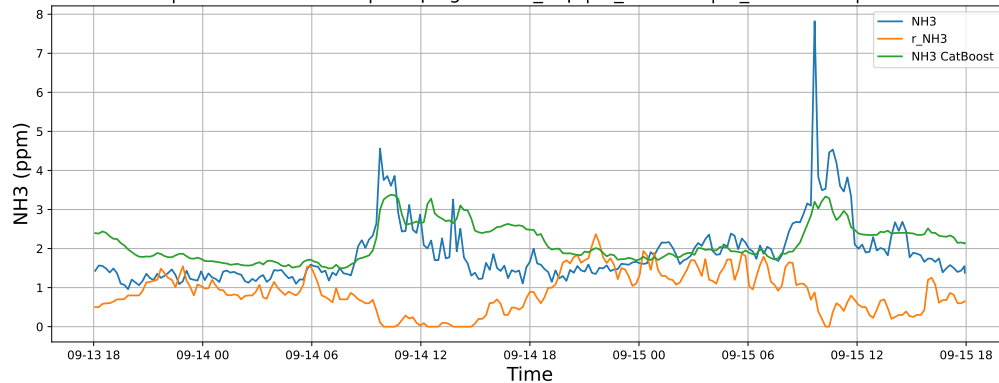
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-13 → 2025-09-15

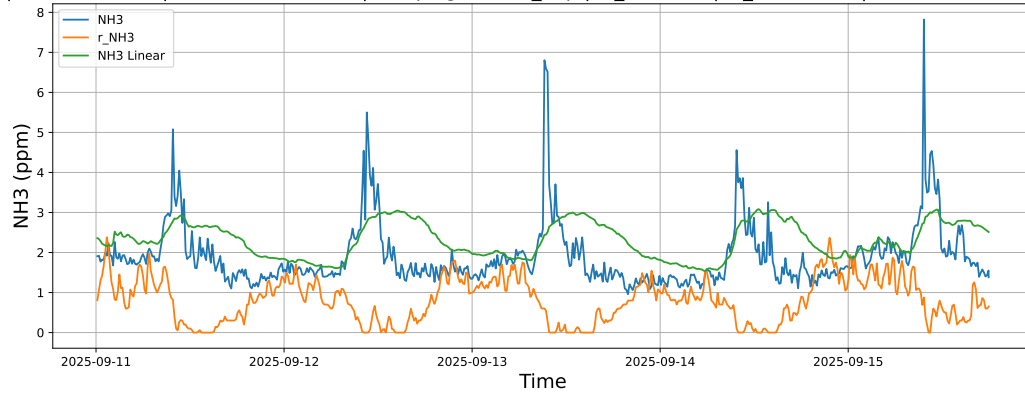


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-13 → 2025-09-15

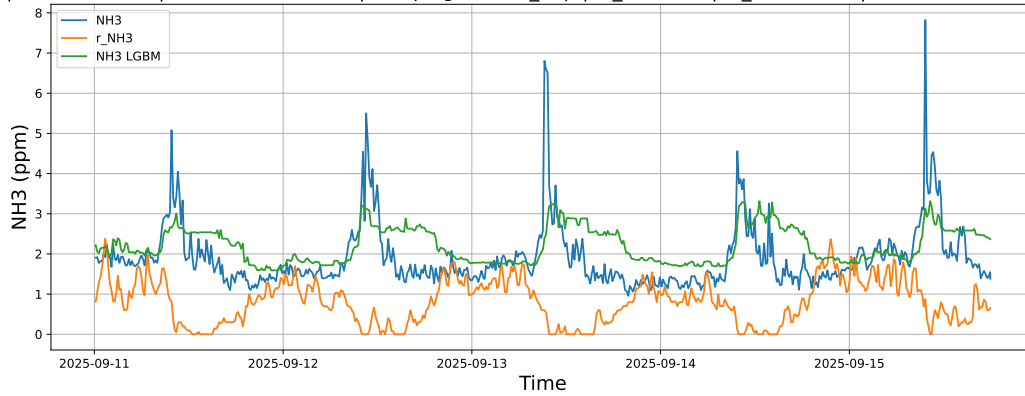


### 3.1.6.3 Time window : ALL

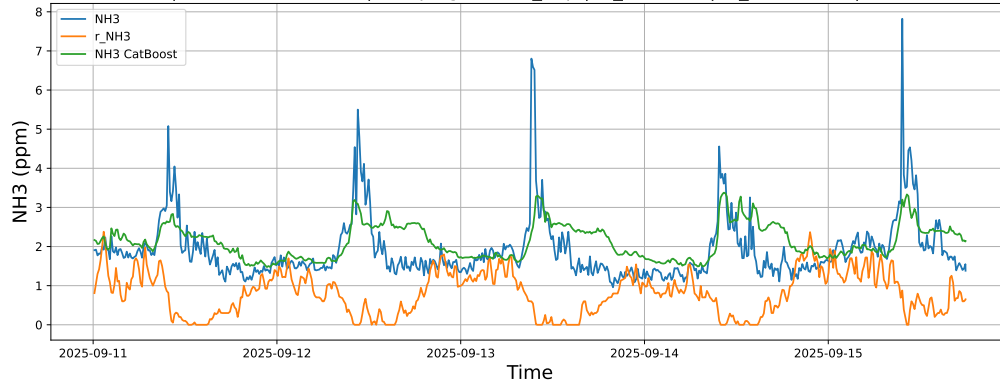
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-11 → 2025-09-15



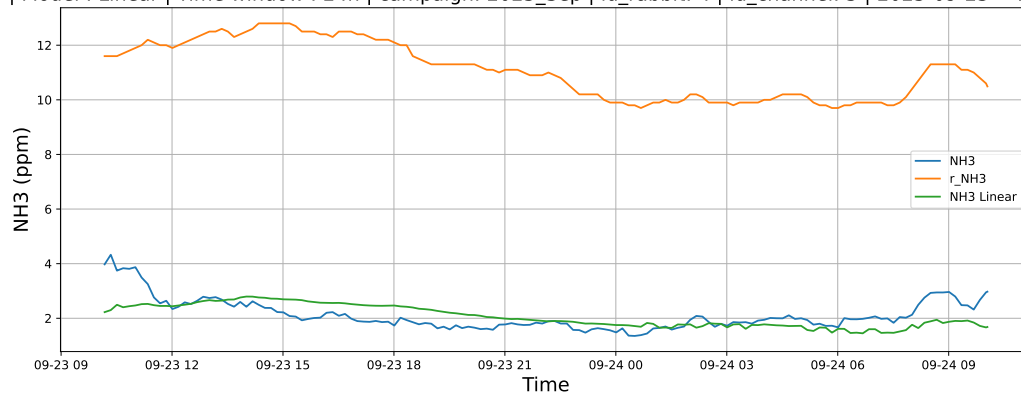
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-11 → 2025-09-15



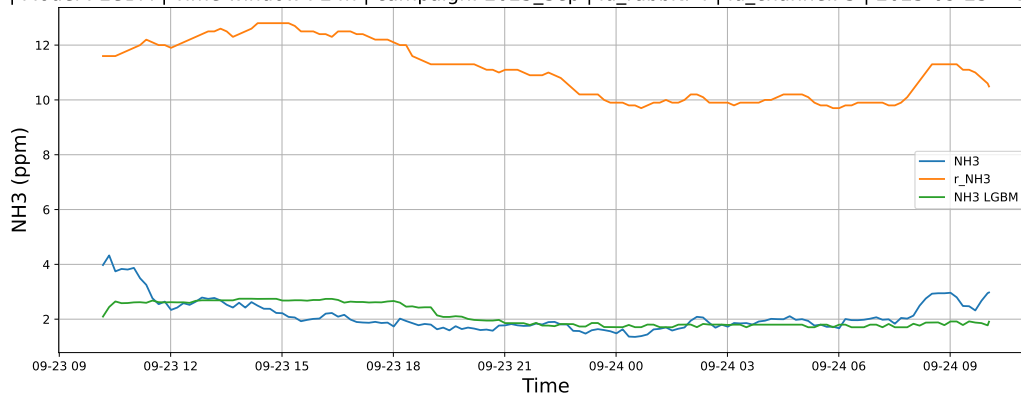
### 3.1.7 Group: campaign: 2025\_Sep,id\_rabbit: 4,id\_channel: 5

#### 3.1.7.1 Time window : 24

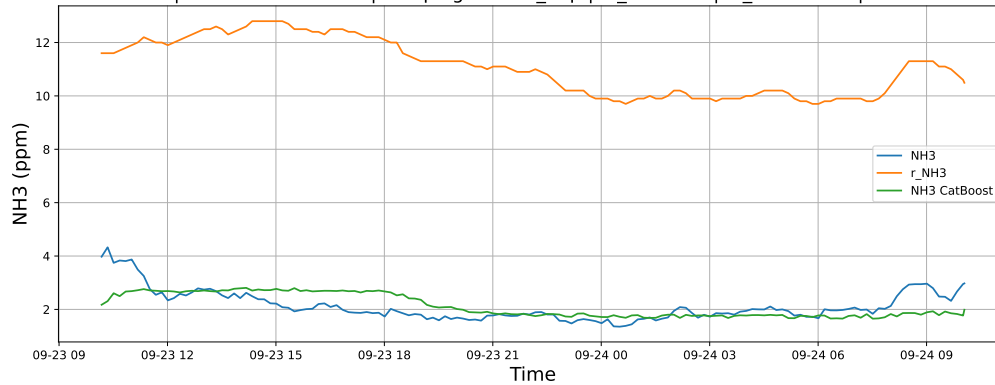
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-23 → 2025-09-24

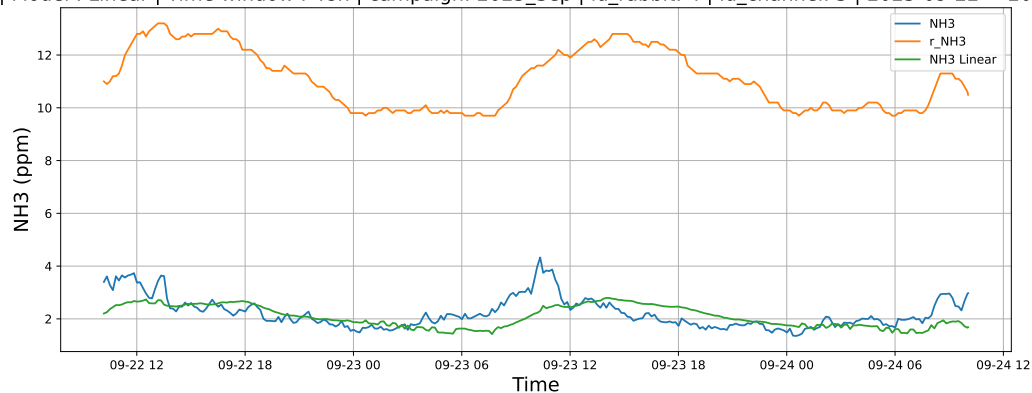


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-23 → 2025-09-24

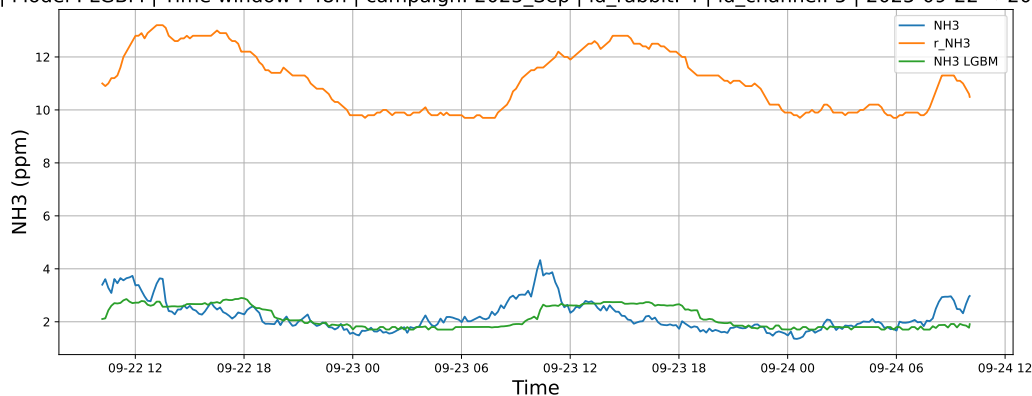


### 3.1.7.2 Time window : 48

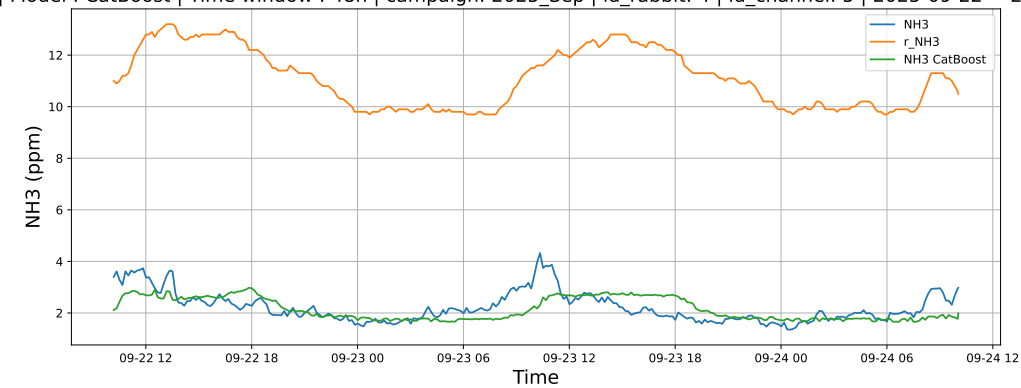
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-22 → 2025-09-24



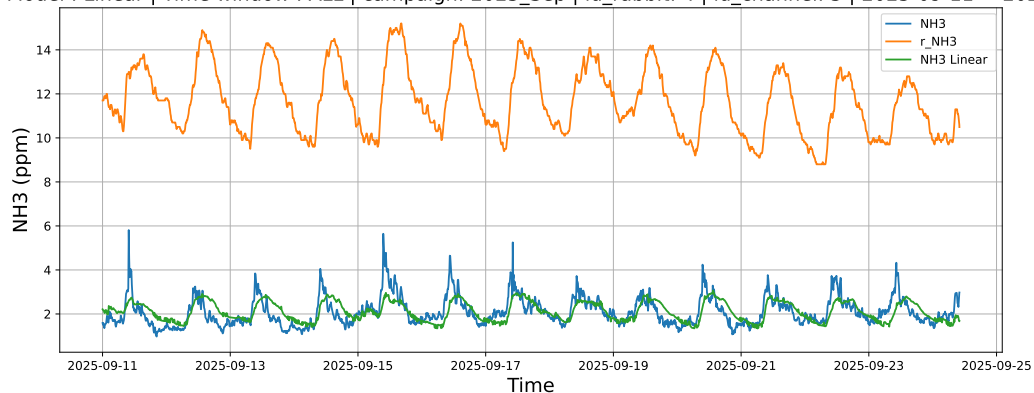
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-22 → 2025-09-24



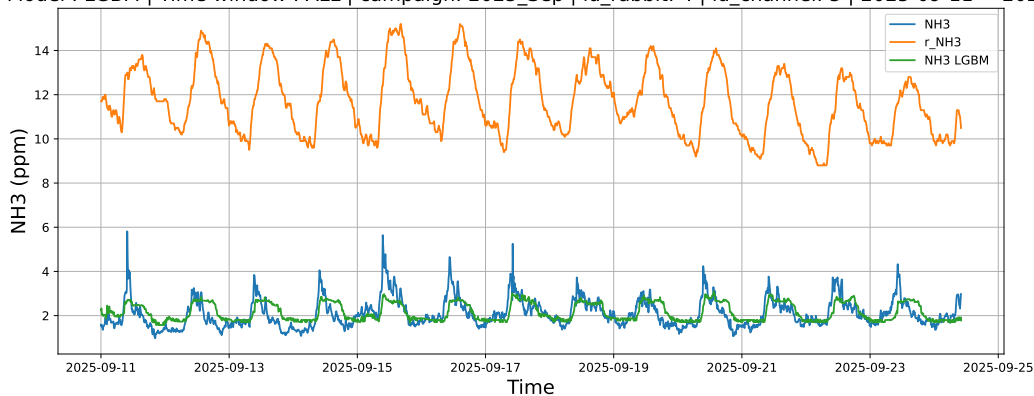
### 3.1.7.3 Time window : ALL



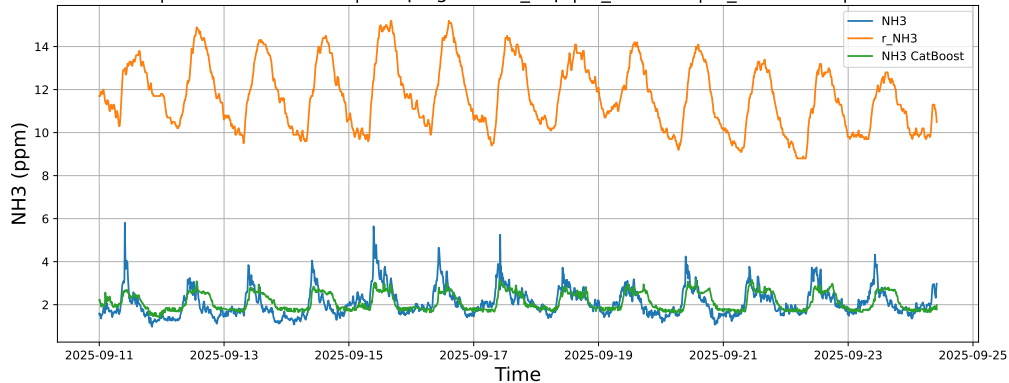
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-11 → 2025-09-24



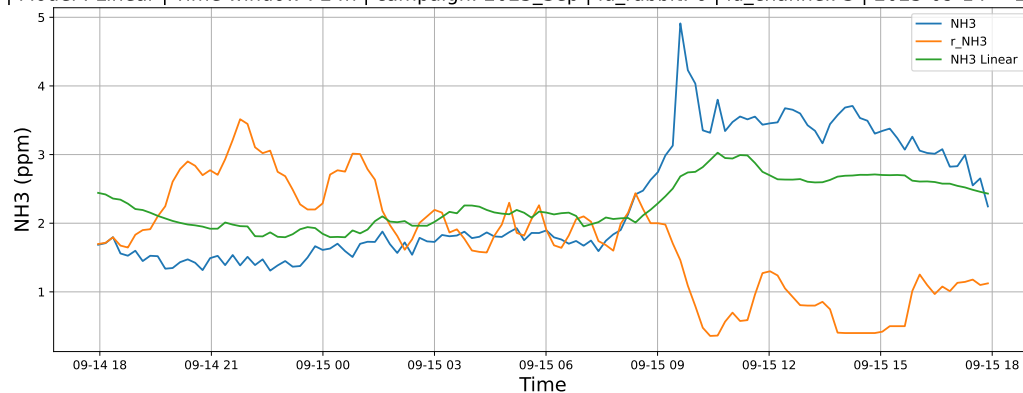
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-11 → 2025-09-24



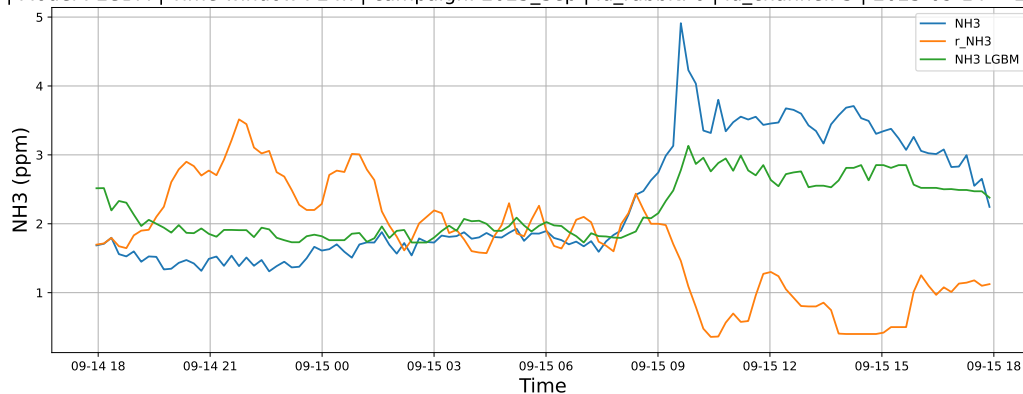
### 3.1.8 Group: campaign: 2025\_Sep,id\_rabbit: 6,id\_channel: 3

#### 3.1.8.1 Time window : 24

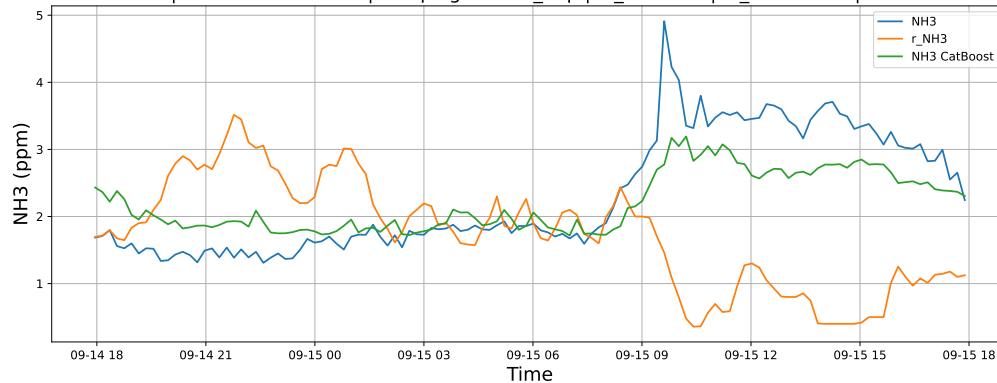
NH3 | Model : Linear | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-14 → 2025-09-15

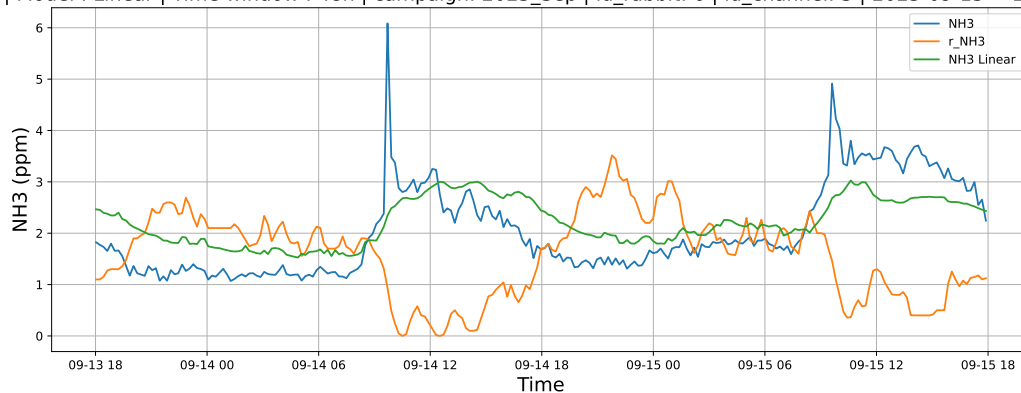


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-14 → 2025-09-15

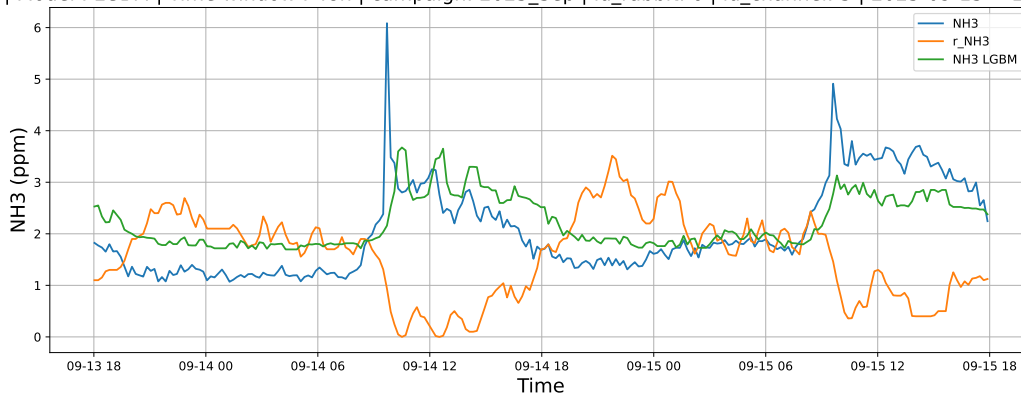


### 3.1.8.2 Time window : 48

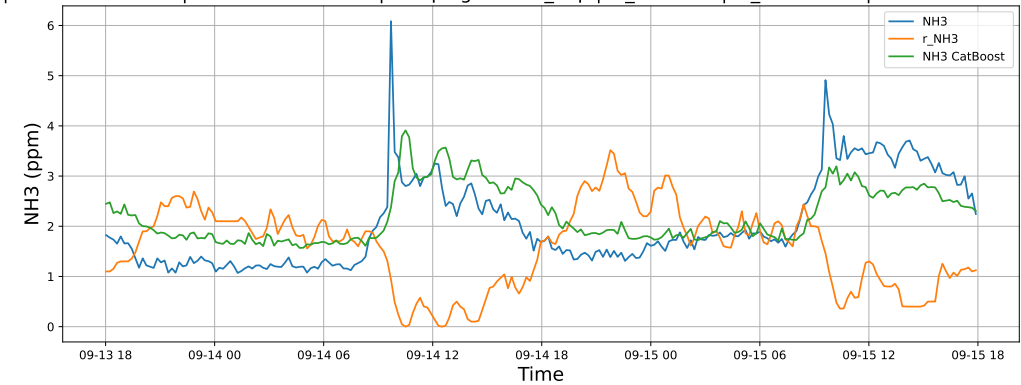
NH3 | Model : Linear | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-13 → 2025-09-15

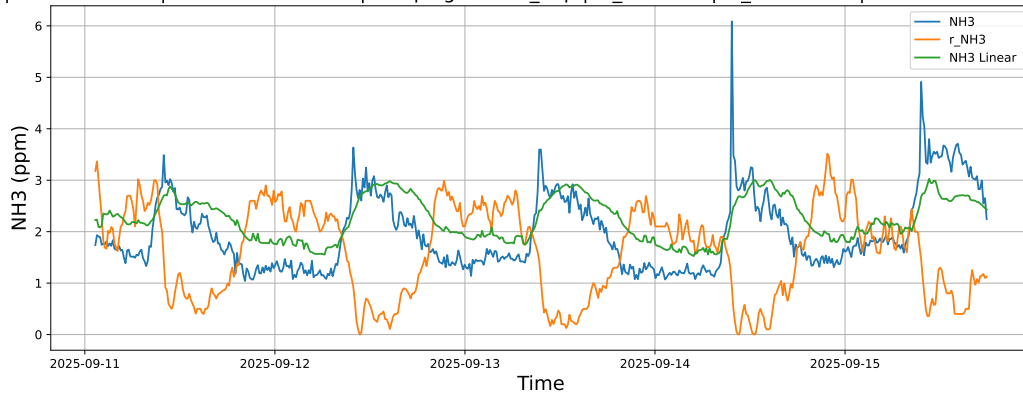


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-13 → 2025-09-15

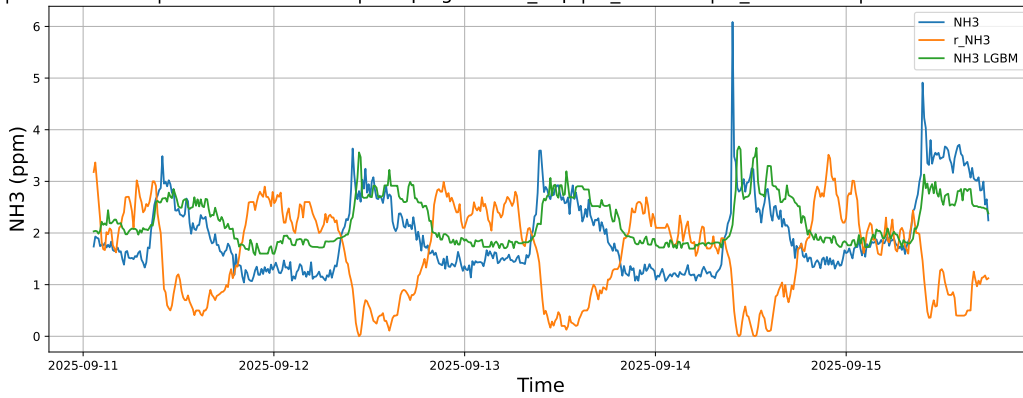


### 3.1.8.3 Time window : ALL

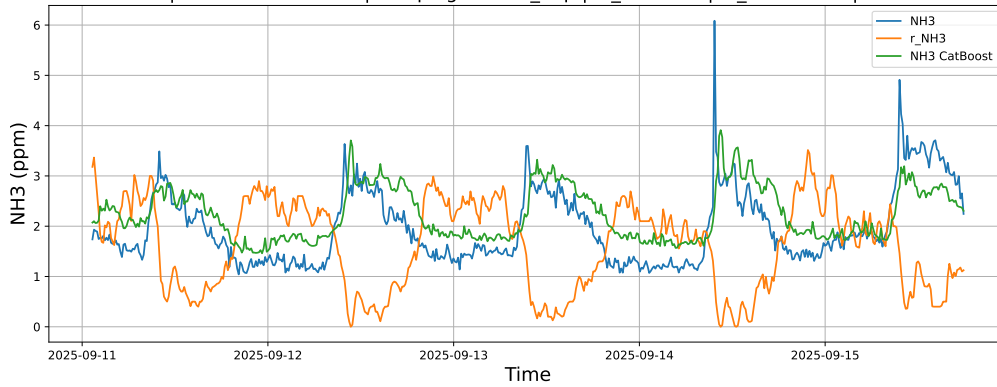
NH3 | Model : Linear | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-11 → 2025-09-15

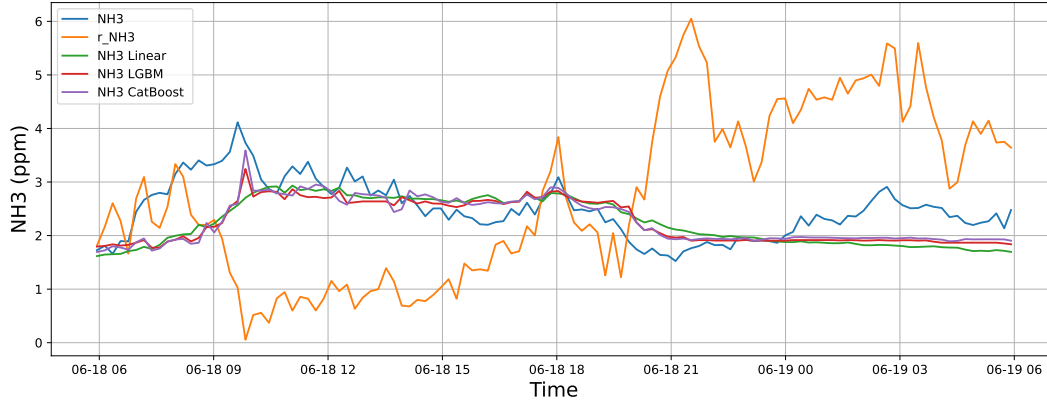


## 3.2 Compare plots

### 3.2.1 Group: campaign: 2025\_Jun,id\_rabbit: 1,id\_channel: 3

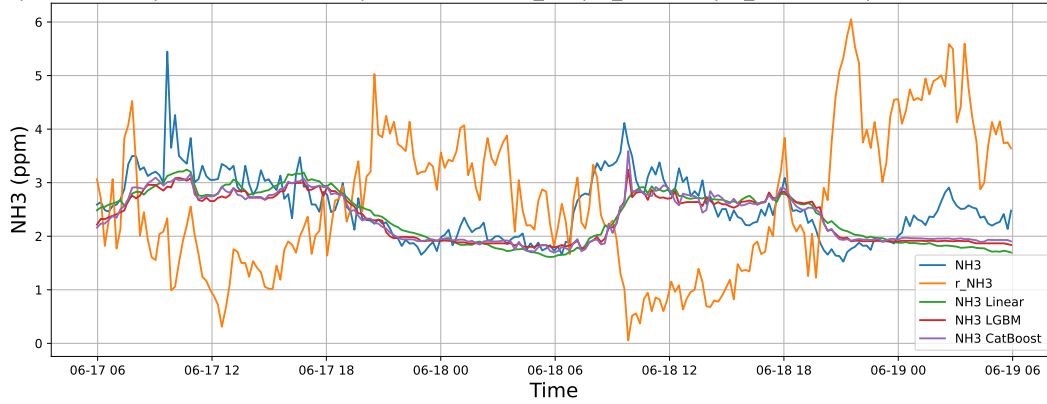
#### 3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-18 → 2025-06-19



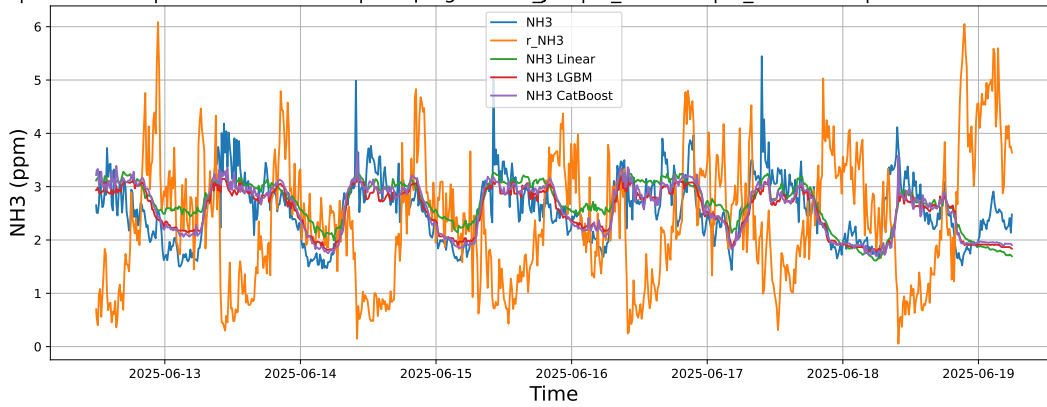
### 3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-17 → 2025-06-19



### 3.2.1.3 Time window : ALL

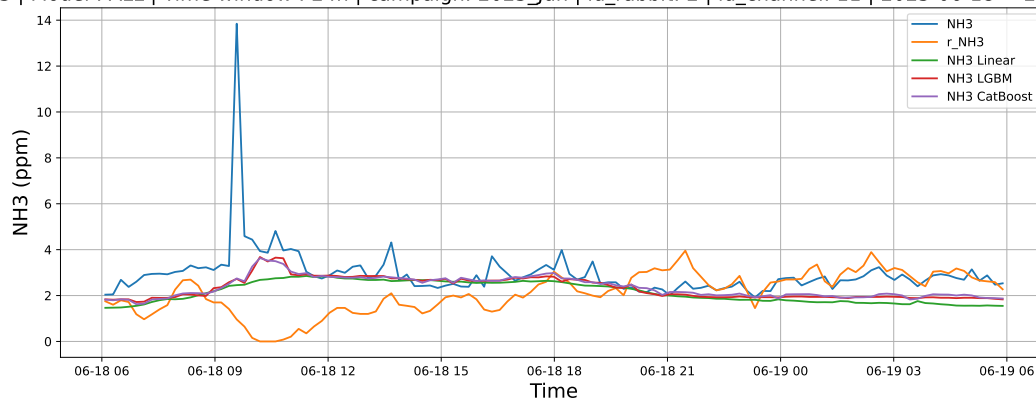
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 1 | id\_channel: 3 | 2025-06-12 → 2025-06-19



### 3.2.2 Group: campaign: 2025\_Jun,id\_rabbit: 2,id\_channel: 11

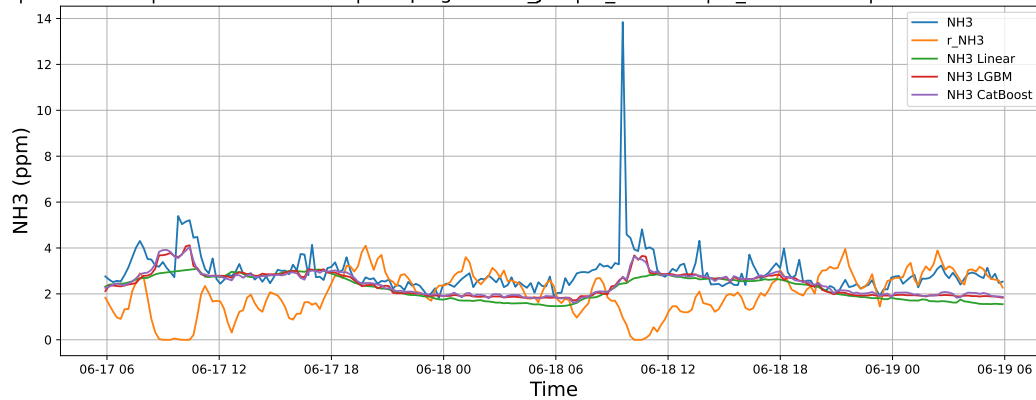
#### 3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-18 → 2025-06-19



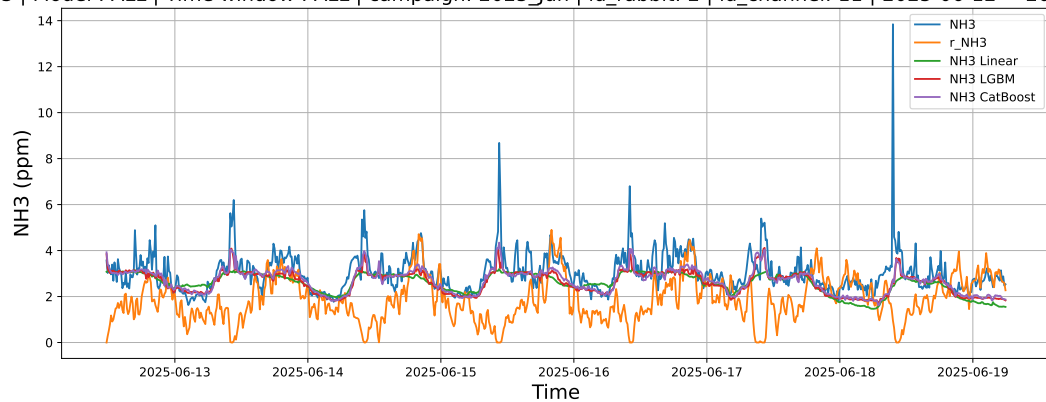
#### 3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-17 → 2025-06-19



#### 3.2.2.3 Time window : ALL

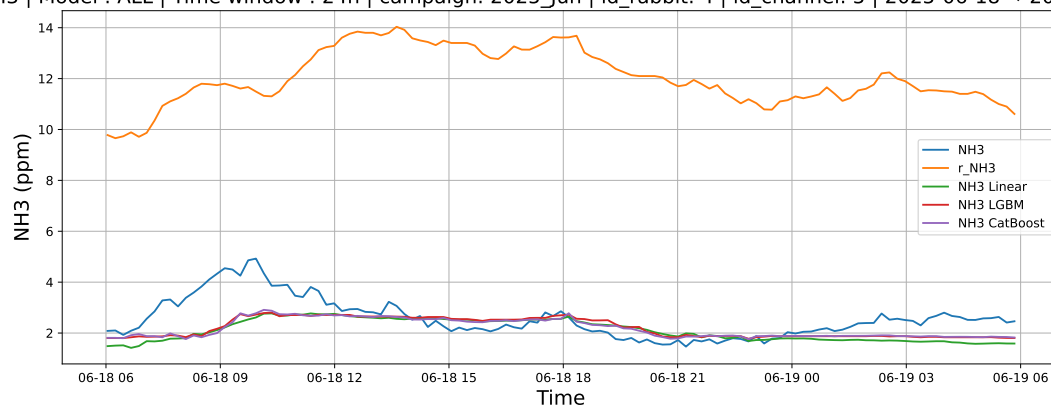
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 2 | id\_channel: 11 | 2025-06-12 → 2025-06-19



### 3.2.3 Group: campaign: 2025\_Jun,id\_rabbit: 4,id\_channel: 5

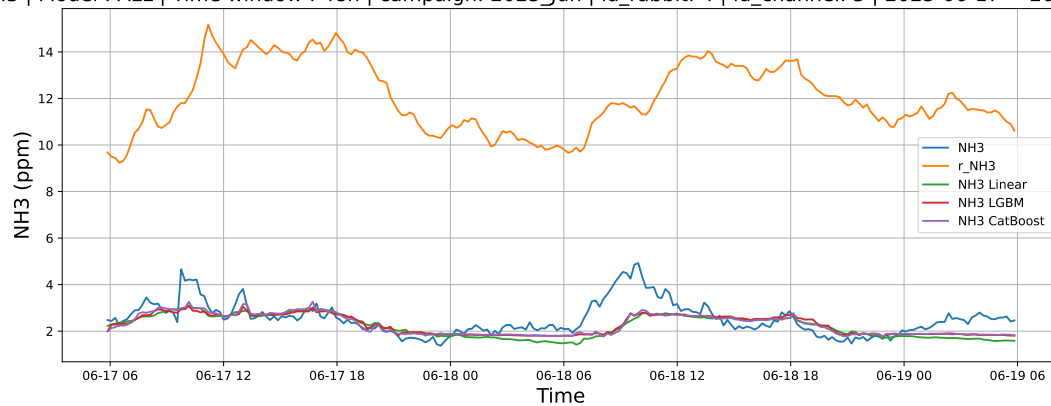
#### 3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-18 → 2025-06-19



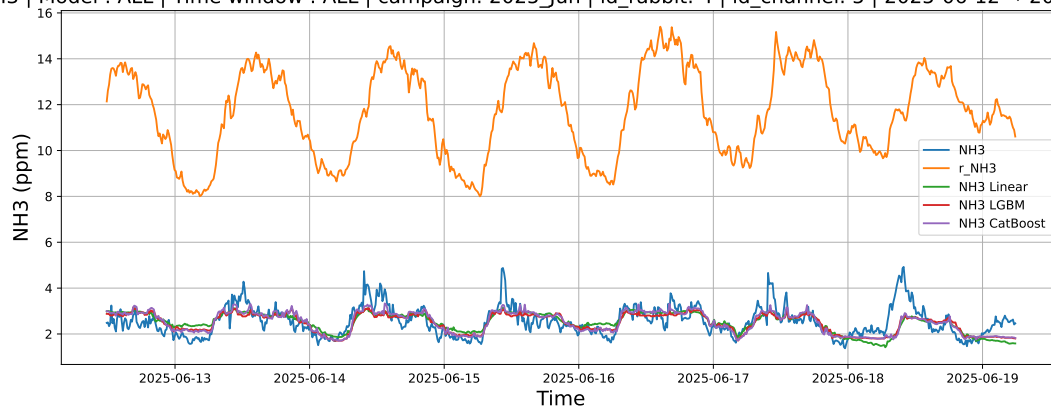
#### 3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-17 → 2025-06-19



### 3.2.3.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 4 | id\_channel: 5 | 2025-06-12 → 2025-06-19

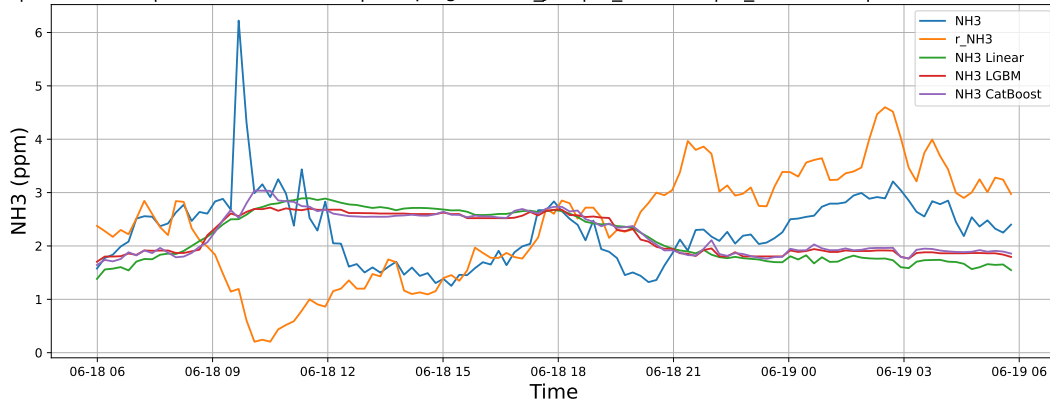


### 3.2.4 Group: campaign: 2025\_Jun,id\_rabbit: 6,id\_channel: 4

#### 3.2.4.1 Time window : 24

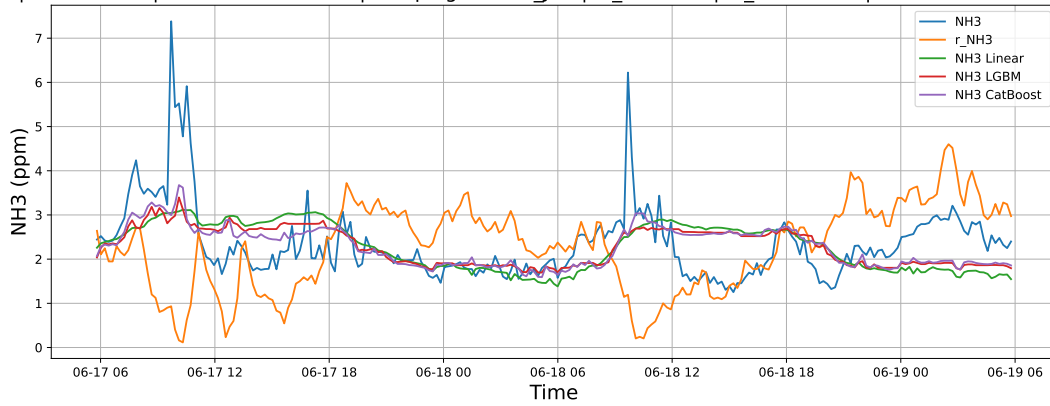


NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-18 → 2025-06-19



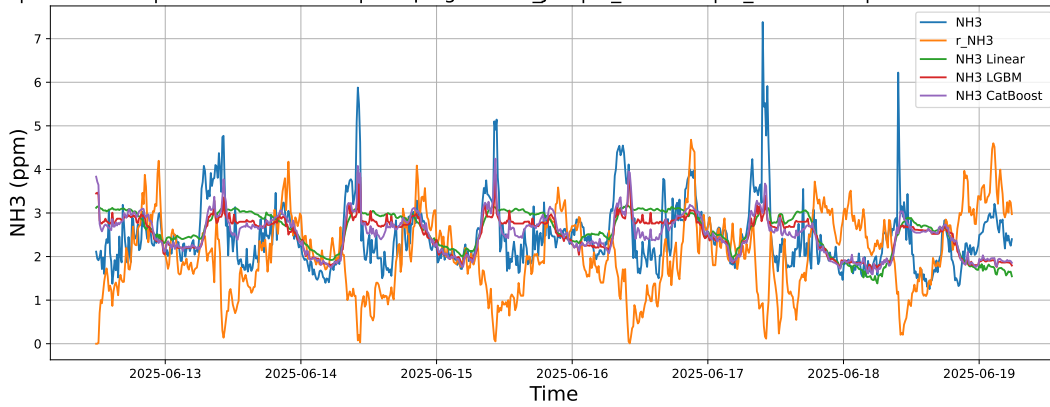
### 3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-17 → 2025-06-19



### 3.2.4.3 Time window : ALL

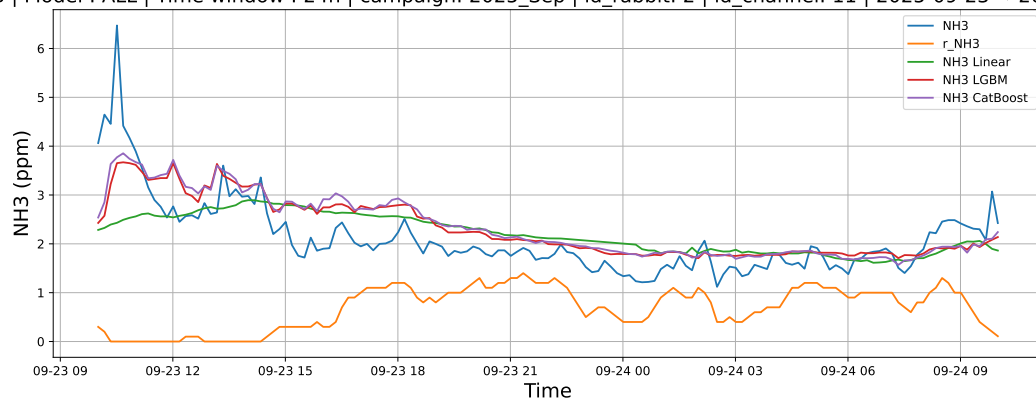
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Jun | id\_rabbit: 6 | id\_channel: 4 | 2025-06-12 → 2025-06-19



### 3.2.5 Group: campaign: 2025\_Sep,id\_rabbit: 2,id\_channel: 11

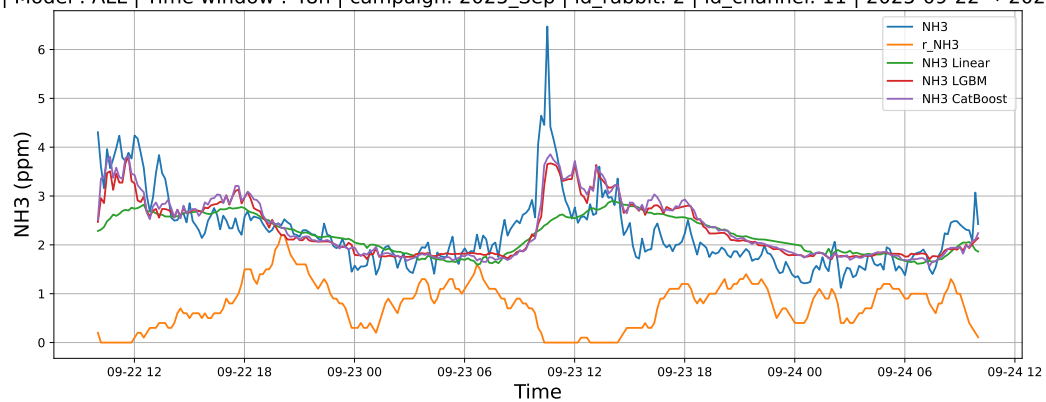
#### 3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-23 → 2025-09-24



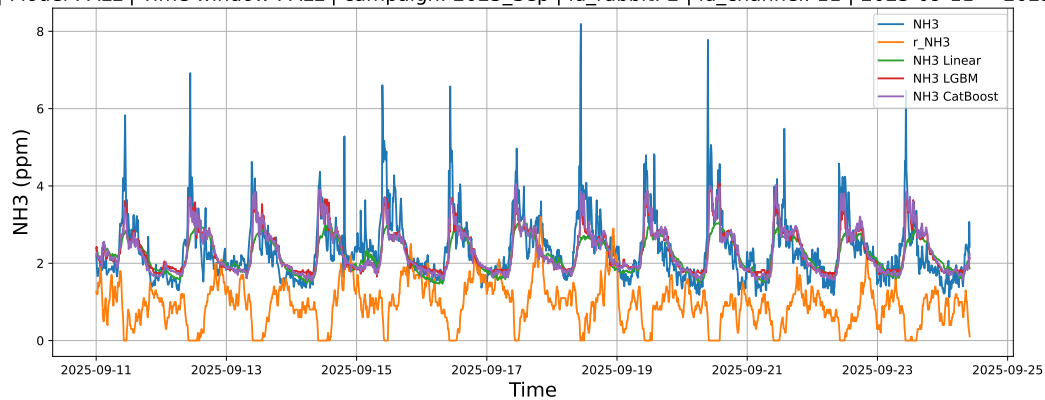
#### 3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-22 → 2025-09-24



#### 3.2.5.3 Time window : ALL

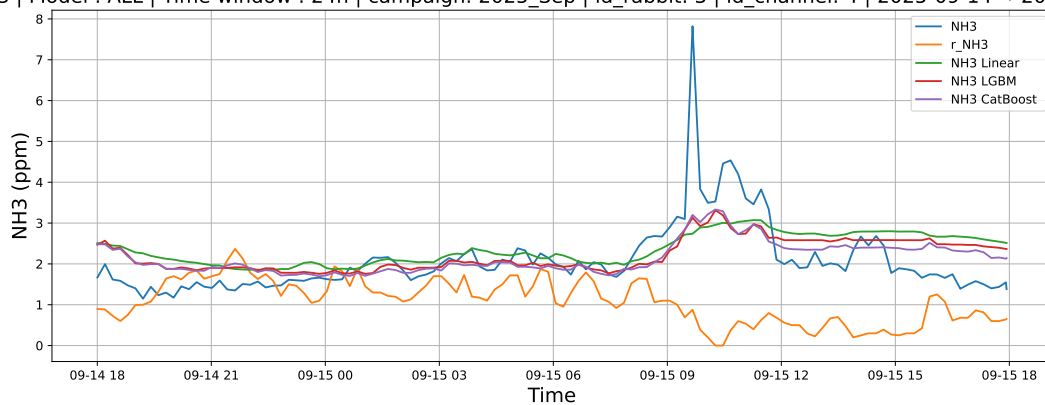
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 2 | id\_channel: 11 | 2025-09-11 → 2025-09-24



### 3.2.6 Group: campaign: 2025\_Sep,id\_rabbit: 3,id\_channel: 4

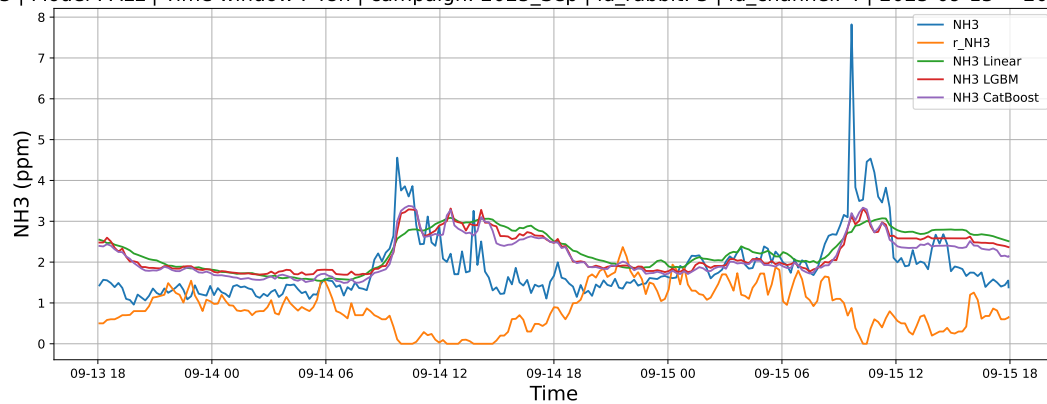
#### 3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-14 → 2025-09-15



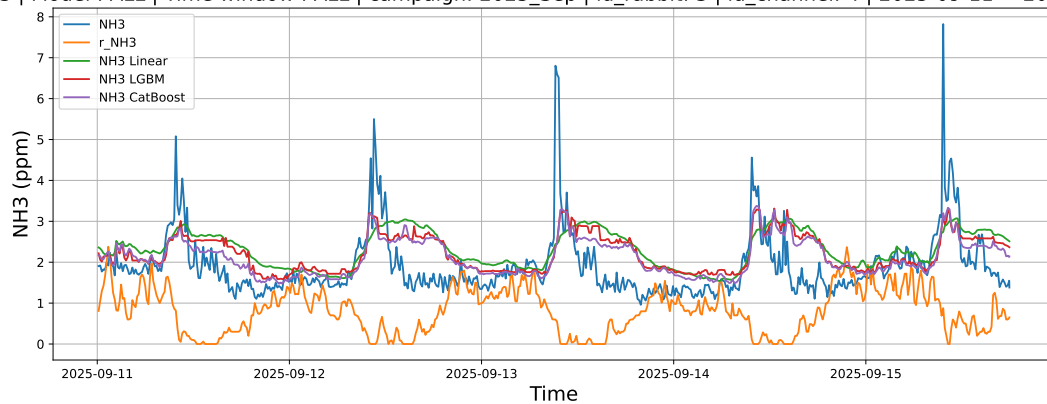
#### 3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-13 → 2025-09-15



### 3.2.6.3 Time window : ALL

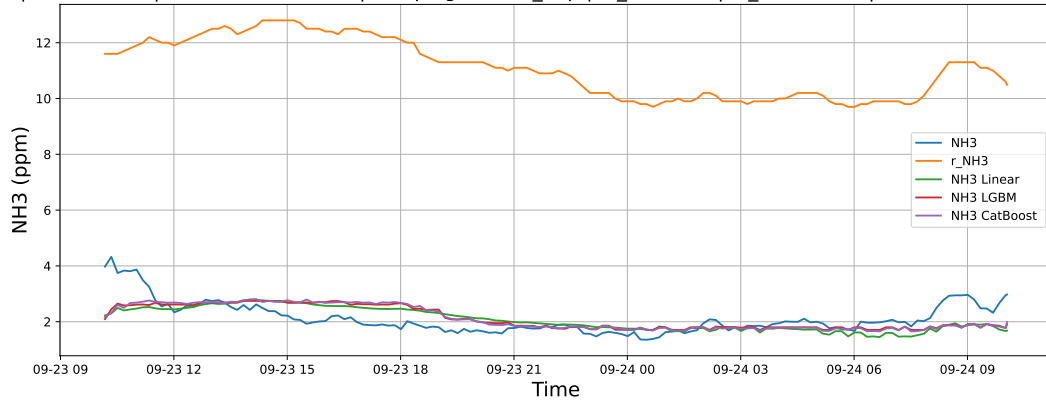
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 3 | id\_channel: 4 | 2025-09-11 → 2025-09-15



### 3.2.7 Group: campaign: 2025\_Sep,id\_rabbit: 4,id\_channel: 5

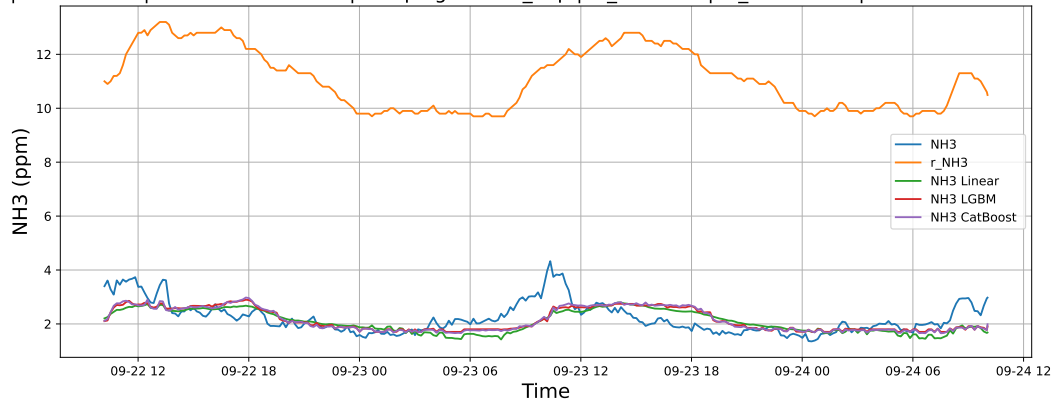
#### 3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-23 → 2025-09-24



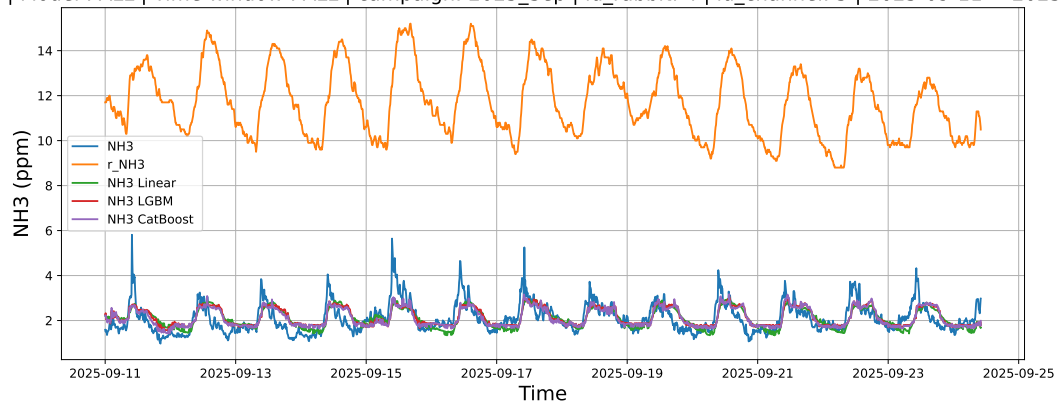
### 3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-22 → 2025-09-24



### 3.2.7.3 Time window : ALL

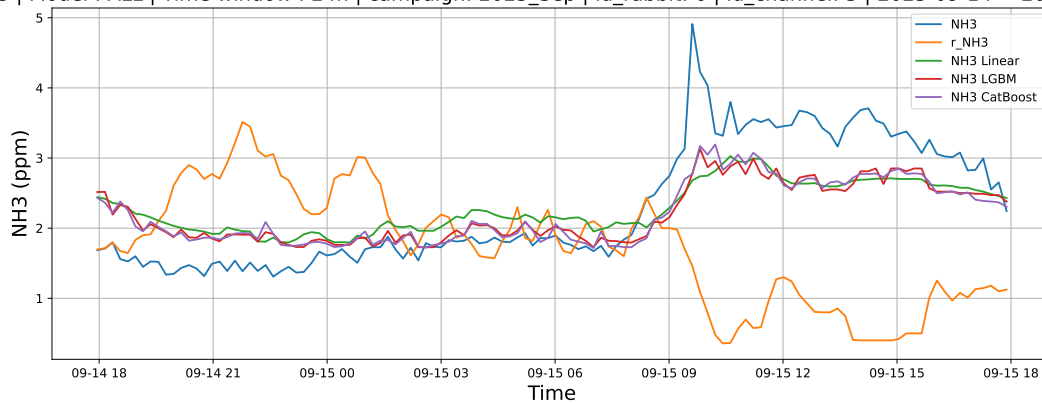
NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 4 | id\_channel: 5 | 2025-09-11 → 2025-09-24



### 3.2.8 Group: campaign: 2025\_Sep,id\_rabbit: 6,id\_channel: 3

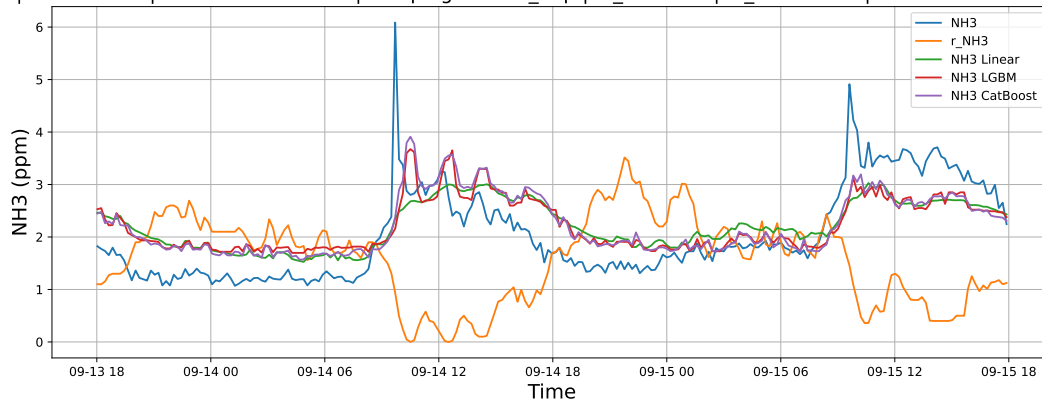
#### 3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-14 → 2025-09-15



#### 3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-13 → 2025-09-15



#### 3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025\_Sep | id\_rabbit: 6 | id\_channel: 3 | 2025-09-11 → 2025-09-15

